

Network Fundamentals

CCNA 200-301 V1.1

MODULE 1

Topics

- 1.1 Intro to Networking
- 1.2 OSI Model
- 1.3 TCP/IP
- 1.4 Network Components
- 1.5 Network Topology
- 1.6 Copper Cable Types
- 1.7 Fiber Optic Cable Types
- 1.8 Virtualization Fundamentals
- 1.9 Review



1.1 Intro to Networking

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Module 1

Basic Network Concepts

OSI Model

TCP/IP Model

TCP, UDP, IP

ICMP, IGMP, ARP

What is a Network?

Two or more computers (nodes) connected together

The computers can be any type of computing device

The connection can be wired or wireless

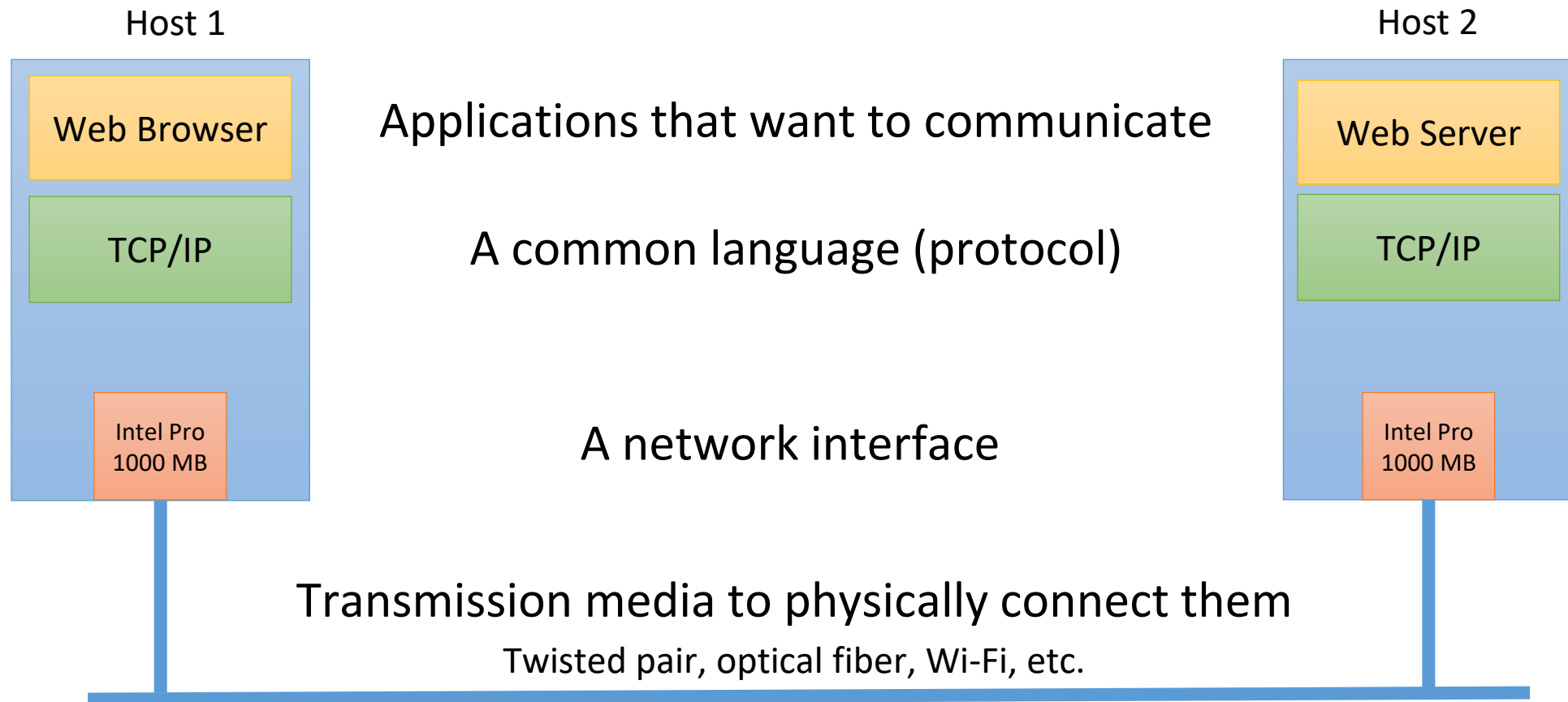
Use cases for a network include:

- Share data
- Remote communication
- Share resources such as printers, faxes, databases, and services
- Distribute a computing workload

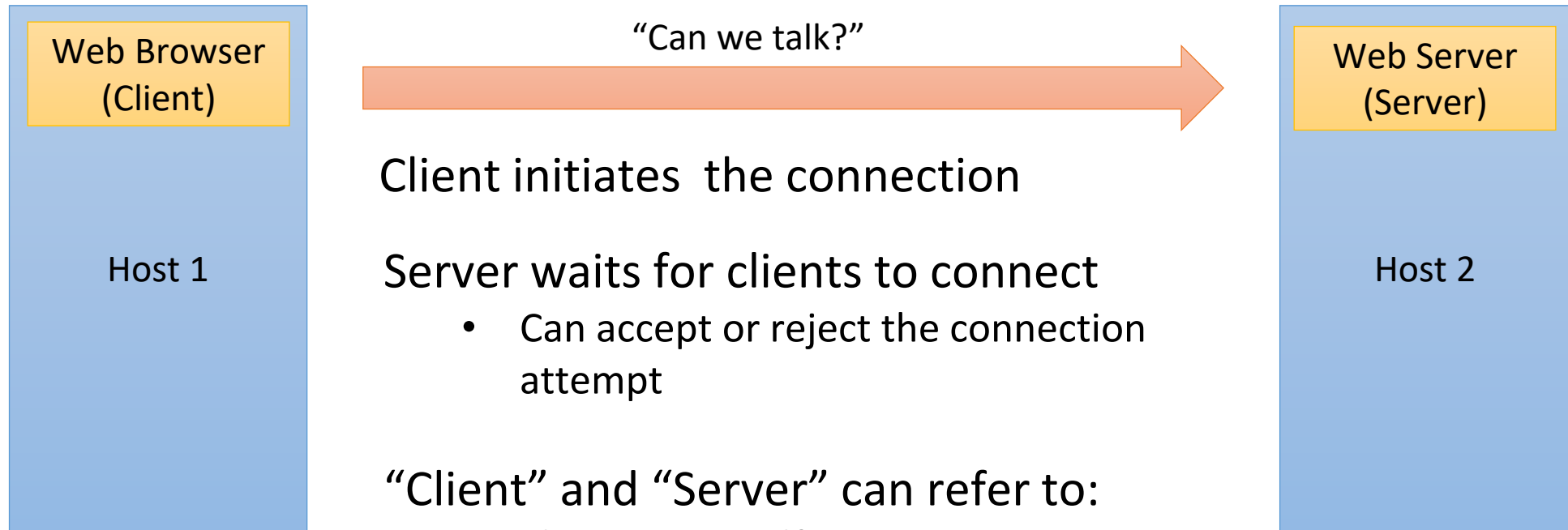
“Wired” can be copper-based or optical fiber



Networking Components



Client / Server



“Client” and “Server” can refer to:

- The device itself
- A process or service running on the device



What is an IP Address?

A logical number that identifies a node on the network

- Analogous to a phone number

Can be changed

- Manually configured in network settings
- Device automatically obtains a new address when connected to a different network

A device can have more than one IP address

- One per interface
- Alternate IP addresses on the same interface

Each IP address must be unique on the entire network so there is no conflict

Examples:

IPv4 - 192.168.1.10

IPv6 - 2601:140:8780:43f0:2d1e:9ebb:92f0:d6e9



What is a MAC Address?

The physical address of the network interface

- AKA “Burned In Address” (BIA)
- 12 hexadecimal numbers
- Does not change

Assigned by the NIC vendor

One per interface

Analogous to a cell phone MEID or IMEI number, or a device serial number

Used in Ethernet, Wi-Fi, and Token Ring networks

MAC addresses must be unique on a network segment

Examples: BC-85-56-F3-13-02, BC:85:56:F3:13:02, BC85.56F3.1302

Windows

Linux, macOS

Cisco



Relationship Between IP and MAC Addresses

IP addresses stay constant end-to-end

- They represent the final source and destination node addresses
- They do not change, regardless of how far away the nodes are from each other

MAC addresses change with each hop (network segment)

- They represent the next node the packet must be handed to
- The nodes relay the packet from one node to the next until it reaches its final destination

IP 10.1.1.1



MAC abab

MAC bcbc



MAC 1212

MAC 3434



MAC cdcd

MAC dede



MAC 5656

MAC 7878

IP 20.2.2.2



With each hop along the path, the source and destination MAC change
The source and destination IP address stay the same



What is a Protocol?

Set of rules or “language” for communication

Protocols exist at every level/layer of networking

A host will use multiple protocols to make a connection on the network



What is a Port?

A number that represents an application (process) on the network

Services use well-known ports

Client apps have their ports temporarily assigned by the operating system

Every process that accesses the network has its own unique port number on that device

A single device can have many running processes, each with its own port



What is a Socket?

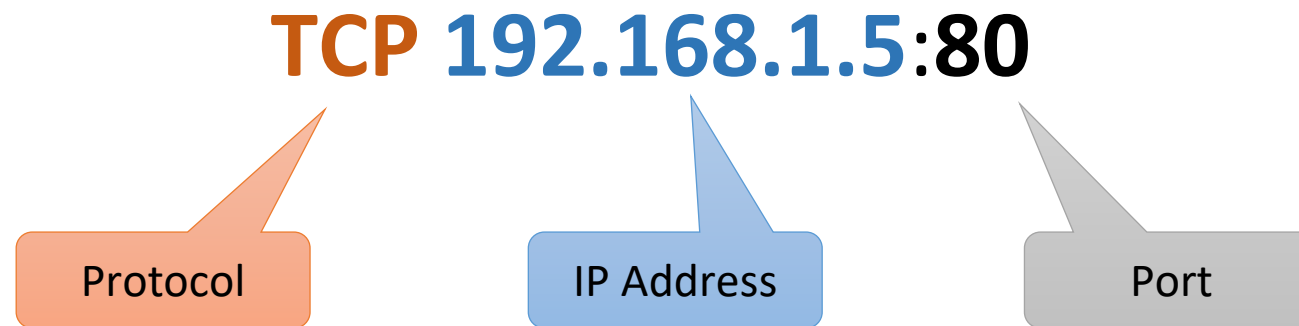
A socket is a port that is in use

It is a combination of protocol, IP address, and port

This combination uniquely identifies the connection on the network

- You cannot have two processes trying to use the same socket at the same time

Example:



What are Source and Destination?

Source is the node that is transmitting/sending

Destination is the intended recipient of the transmission

A Client and Server take turns transmitting to each other

- They constantly alternate between being source and destination

Source and destination is used at multiple levels:

- IP address
- Port
- MAC address



What is a Unicast, Multicast, Broadcast?

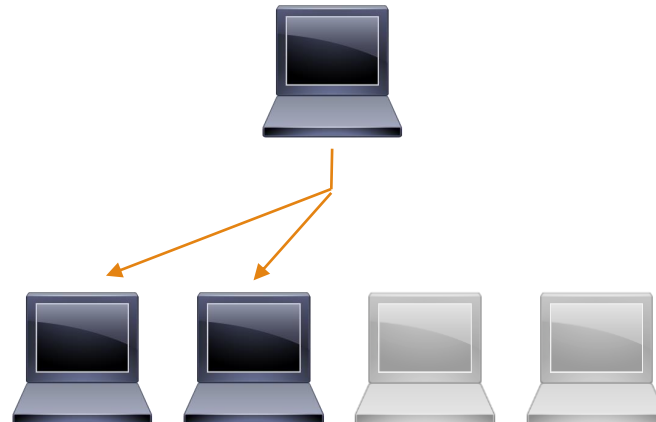
A node transmits a packet on the network

Depending on the destination address, one or more nodes will receive the packet

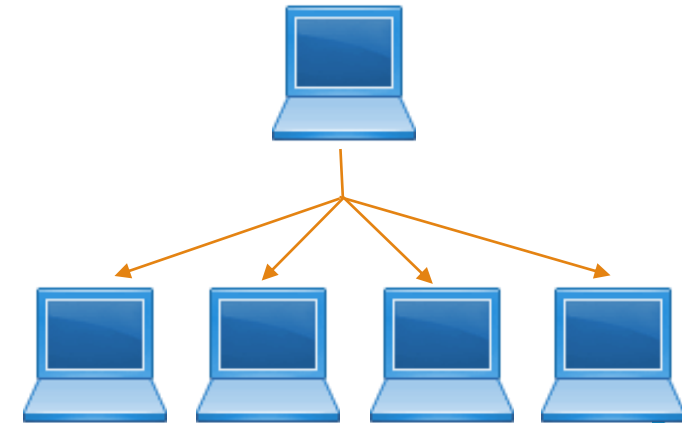
Unicast
One to one



Multicast
One to Some



Broadcast
One to All



Open Systems Interconnection (OSI) Model

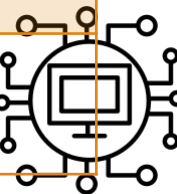
1.2 OSI Model

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Module 1

Open Systems Interconnection (OSI) Model

	Layer	Description	Description
7	Application	Customer service counter for app to request network services	HTTP, HTTPS, FTP, TFTP, SMTP, POP3, IMAP4, SMB, NFS, RDP, LDAP, DNS, DHCP, SSH, Telnet, SNMP ...
6	Presentation	Both sides agree on common data format	All encryption, multimedia formats, character sets, encryption level
5	Session	Keep separate conversations separate	Ports, named pipes, NetBIOS, RPC
4	Transport	Establish, manage, tear down a connection	TCP, UDP
3	Network	Add logical address, choose the best route	IP, ICMP, IGMP
2	Data Link	Format the data for transmission, add physical address	LAN and WAN protocols, ARP
1	Physical	Actually transmit packet as 1's and 0's	All physical and electrical characteristics of connectors and transmission media



1.3 TCP/IP

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Module 1

TCP/IP Model

TCP, UDP, IP

ICMP, IGMP, ARP

TCP/IP Model

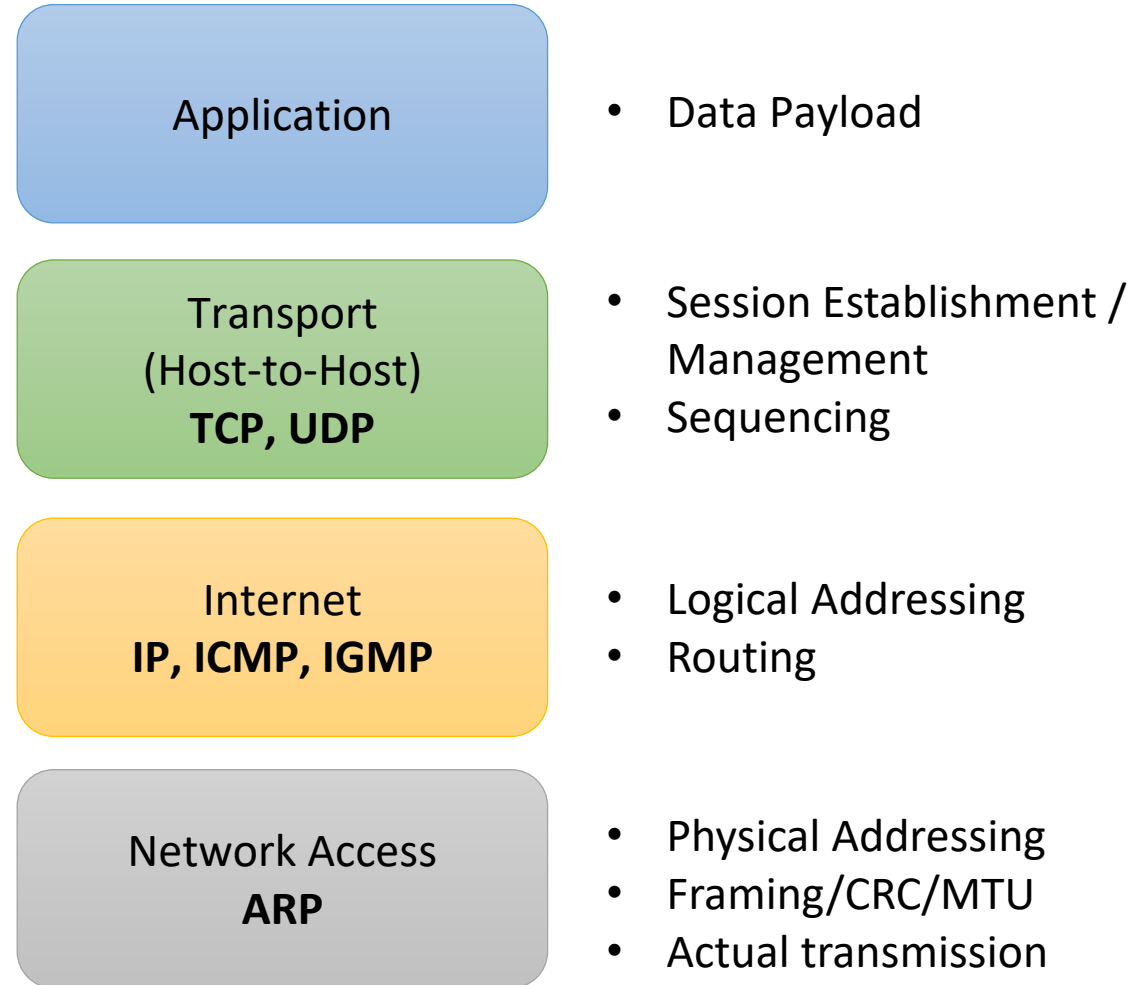
AKA DoD Model

6 core protocols:

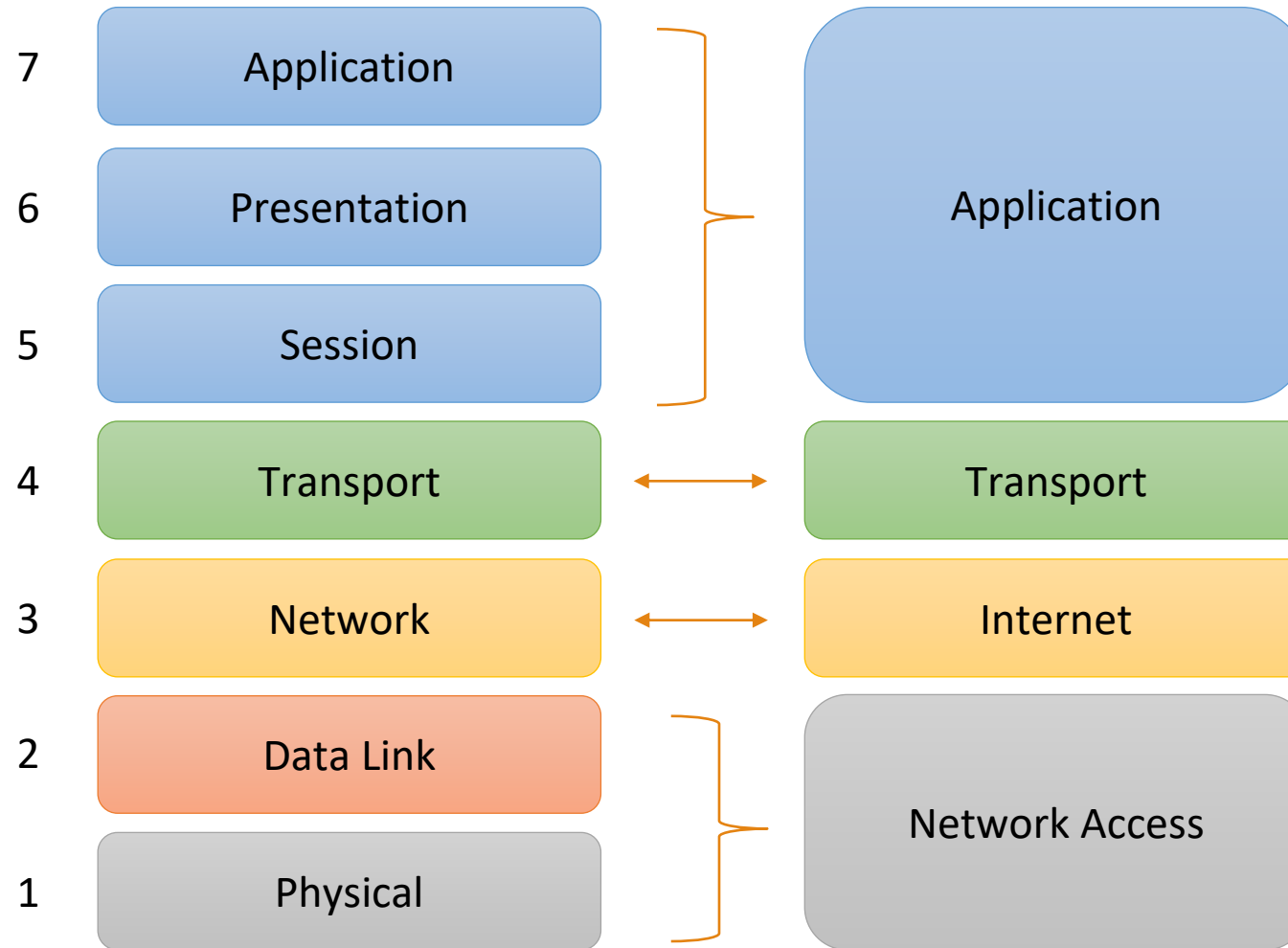
- TCP, UDP, IP, ICMP, IGMP, ARP

All but ARP have an IANA assigned protocol number (protocol ID)

Many auxiliary protocols also exist at the various layers



OSI – TCP/IP Layer Mapping



TCP, UDP, IP



Transmission Control Protocol (TCP)

Layer 4 host-to-host protocol

- Payload of IP
- Protocol ID 6
- Embeds source and destination port in its header

Connection-oriented

- Attempts to guarantee delivery using checksums, acknowledgements, and retransmissions

Breaks down a large file into smaller segments

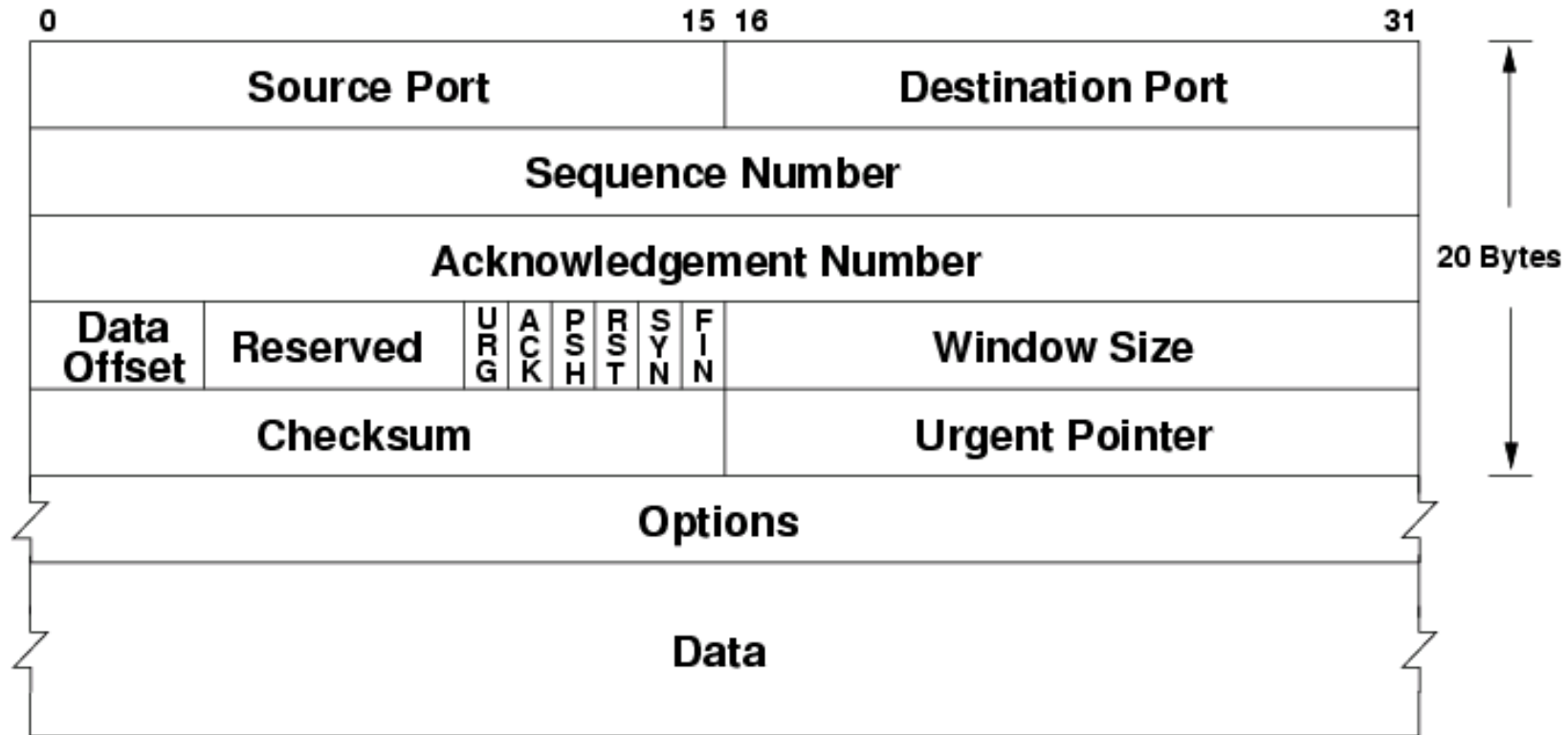
- Each segment identified by sequence #
- Sends a few segments at a time, waits for acknowledgement before sending more

Establishes a connection using a three-way handshake (SYN – SYN/ACK – ACK)

Session closed by a four-way handshake (FIN-ACK-FIN-ACK)



TCP Header



TCP Flags

Indicate the purpose of the TCP segment

Every TCP segment will have at least one flag raised

Flag	Description
URG	Tells receiver to prioritize this data
ACK	Acknowledge All TCP segments (except very first and last will have an ACK)
PSH	Tells receiver to directly send this data to the application
RST	Tells receiver that the sender has abruptly ended the conversation
SYN	Used in 3-way handshake to start conversation
FIN	Used in 4-way handshake to end conversation



User Datagram Protocol (UDP)

TCP's "Little Brother"

Host-to-Host Layer (Layer 4) protocol

- Payload of IP
- Protocol ID 17
- Sends short packets of data, called datagrams
- Embeds source and destination ports in its header

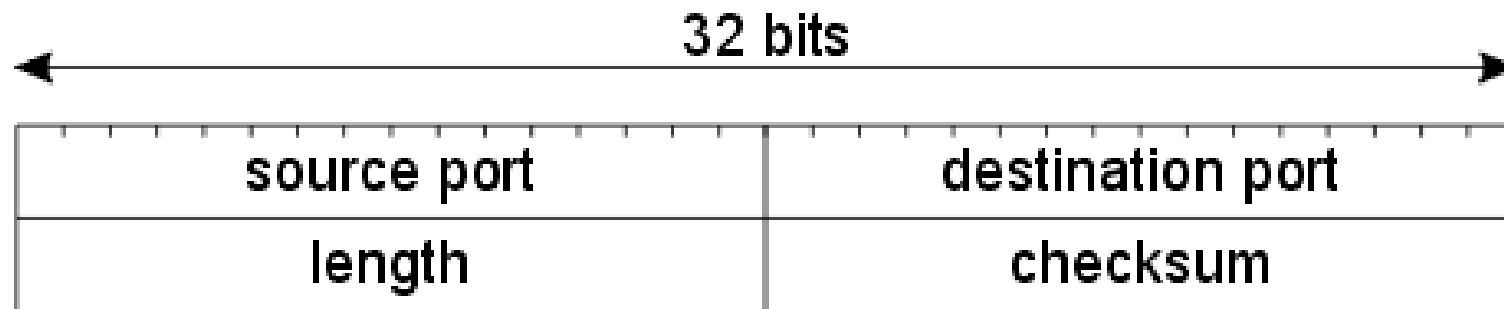
Connectionless ("unreliable" or "best effort")

- Does not guarantee message delivery
- No session establishment
- Uses checksums only
- Depends on the application to provide error correction

A payload of IP



UDP Header



Connection-Oriented vs Connectionless

Connection-Oriented	Connectionless
TCP	UDP, IP, ICMP, IGMP, ARP
Handshake to set up / tear down session	No handshake
Flow control / error correction	No flow control / no error correction
“Reliable”	“Unreliable” or “best effort”
Focus on receiving all the data	Focus on speed/performance
Used where completeness of data transfer is more important than the time it takes: • web pages • emails • file transfers • video-on-demand • remote control	• Used where time is critical but loss is not: • Realtime gaming, voice and video • SNMP, DNS, DHCP



Internet Protocol (IP)

Internet Layer (Layer 3) connectionless protocol

Can be a payload of nearly any link-layer protocol

Protocol ID 4

IPv4:

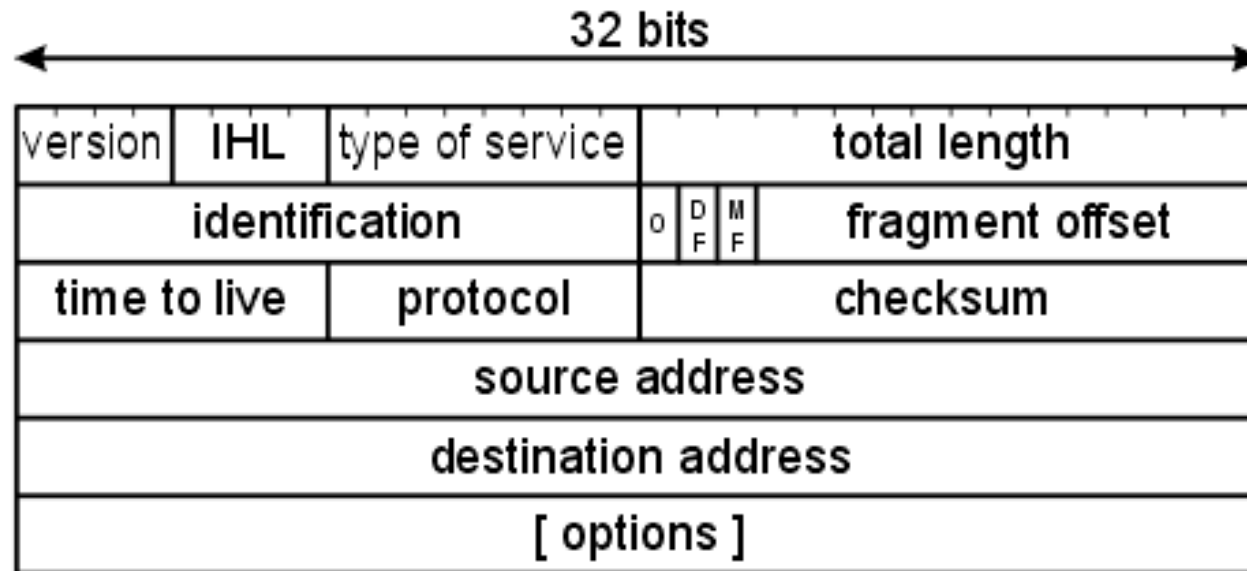
- Still most widely used version
- Uses 32-bit logical addressing to identify source and destination host
- Address examples: 10.0.0.207, 172.16.54.3, 207.45.168.4, 198.32.8.75

IPv6:

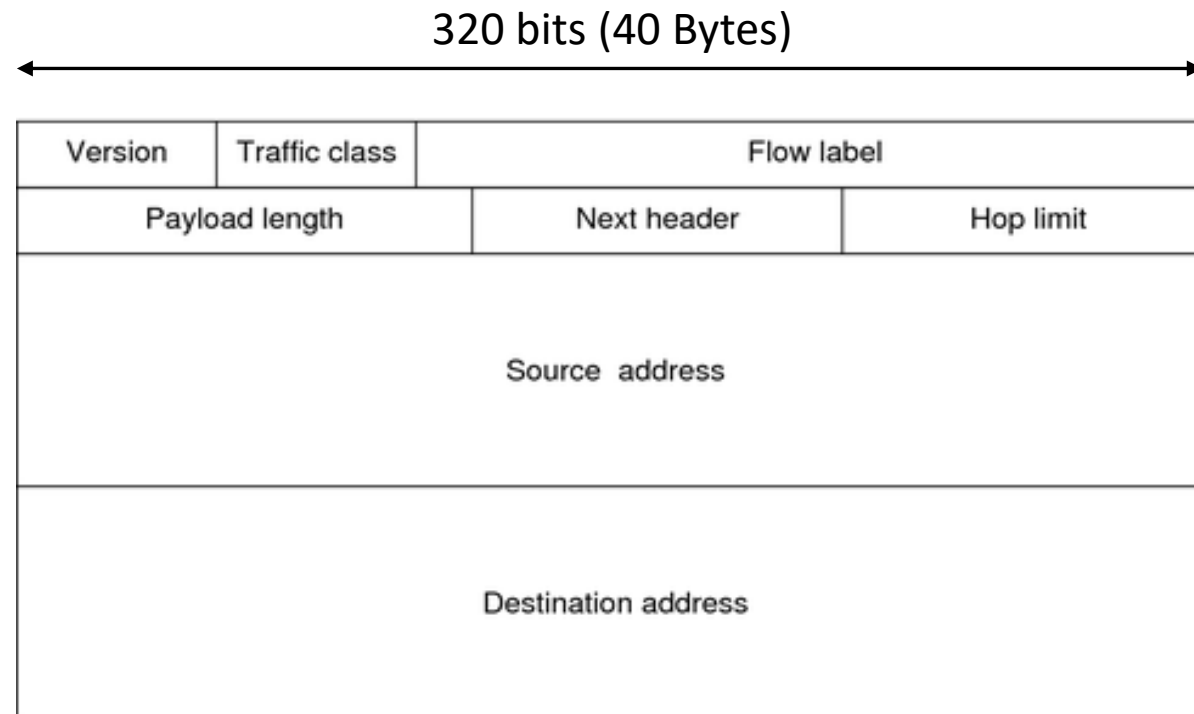
- Allows longer addressing and vast improvements over IPv4
- Steadily gaining in popularity and use
- Address examples: 2601:140:8780:43f0::4762, fe80::10:18ff:fe12:2001



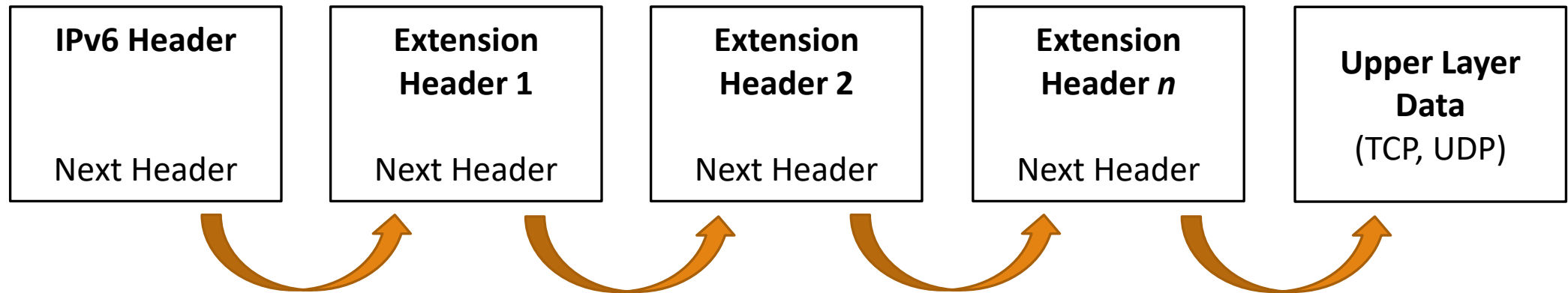
IPv4 Header



IPv6 Header



IPv6 Header Extensions



ICMP, IGMP, ARP



Internet Control Message Protocol (ICMP)

An error-reporting protocol

- Has many message types including echo, echo reply, time exceeded, destination unreachable, redirect

Layer 3 payload of IP

- Protocol ID 1

Used by IP-enabled devices (routers, computers, etc.) and applications such as ping and traceroute



Internet Group Management Protocol (IGMP)

Used by hosts to notify routers that they are still interested in receiving multicasts from an upstream server

- Routers do not forward multicasts by default
- Host must let upstream router know that it wants the multicast

Layer 3 payload of IP

- Protocol ID 2



Address Resolution Protocol (ARP)

Network Access Layer (Layer 2) connectionless protocol

Used to map MAC addresses to IP addresses

Sends Layer 2 broadcast (FFFFFFFFFFFF)

- Queries all listening nodes to identify which one is using the specified IP address

Mappings are temporarily stored in the device's ARP cache

Used in Ethernet and Wi-Fi networks

Does not have an assigned protocol number



1.4 Network Components

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Module 1

Router

Switch

Next-Generation Firewall

Next-Generation IPS

Access Point

Wireless LAN Controller

Endpoint

Server

Power over Ethernet

Router

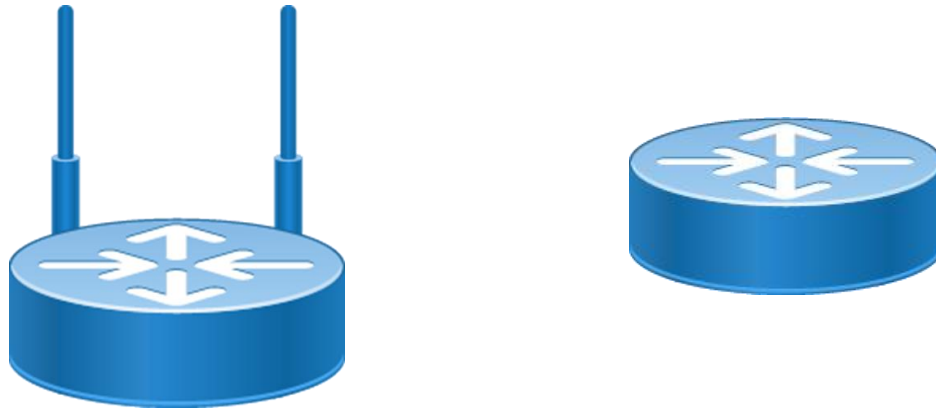
Makes forwarding decisions based on Layer 3 addresses

Maintains a routing table

- Puts the best route to a destination in the table
- Can learn routes from other routers via a routing protocol

Different interfaces allow a router to connect different types of Layer 2 segments

Can have a built-in WAP



Switch

Makes forwarding decisions based on Layer 2 (MAC) addresses

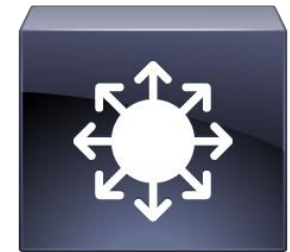
Has high port density

Learns MAC addresses based on frames coming into its ports

- Examines incoming frames
- Puts source MAC addresses into its MAC address table
- Consults MAC table to determine which port to repeat a frame out of

Layer 3 switch also has a router process running in it

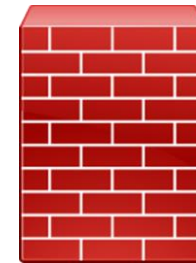
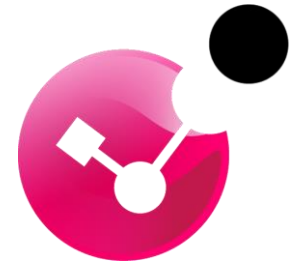
- You can configure a switchport to be an Ethernet router port
- Or you can configure a VLAN interface as a default gateway for its VLAN



Next-Generation Firewall

Firewall with advanced features including:

- VPN server
- Quality-of-Service
- Antivirus scanning
- Deep packet inspection
- SSL/TLS decryption
- Allow/block applications
- Integrated intrusion prevention
- Cloud-based threat intelligence
- Zero-day exploit detection and blocking
- Security policy enforcement based on user, role, group, etc.
- Sandboxing (execute suspicious files and attachments in a virtual environment to detect malicious behavior)

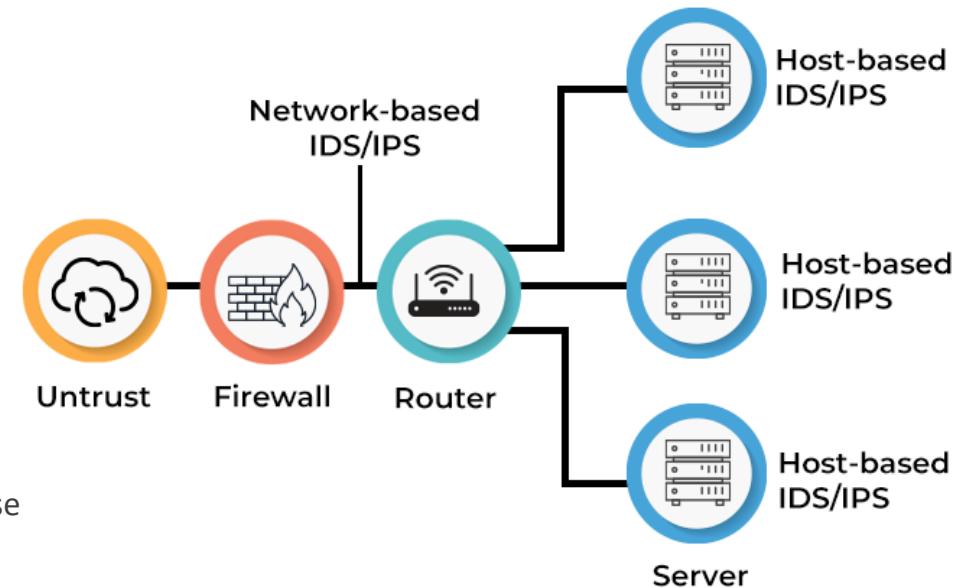


Next-Generation Intrusion Prevention System (IPS)

A network “choke point” between the firewall and the rest of the network

Features include:

- Advanced threat detection
 - Signature, anomaly, and behavior based analysis
- Application awareness
 - Monitor specific apps and their behavior
- Context awareness
 - User identity, device type, location, application behavior
- Threat intelligence integration
 - Auto-updates of the latest threat signatures and indicators of compromise (IoCs)
- Cloud/AI integration
- SSL/TLS decryption



Access Point

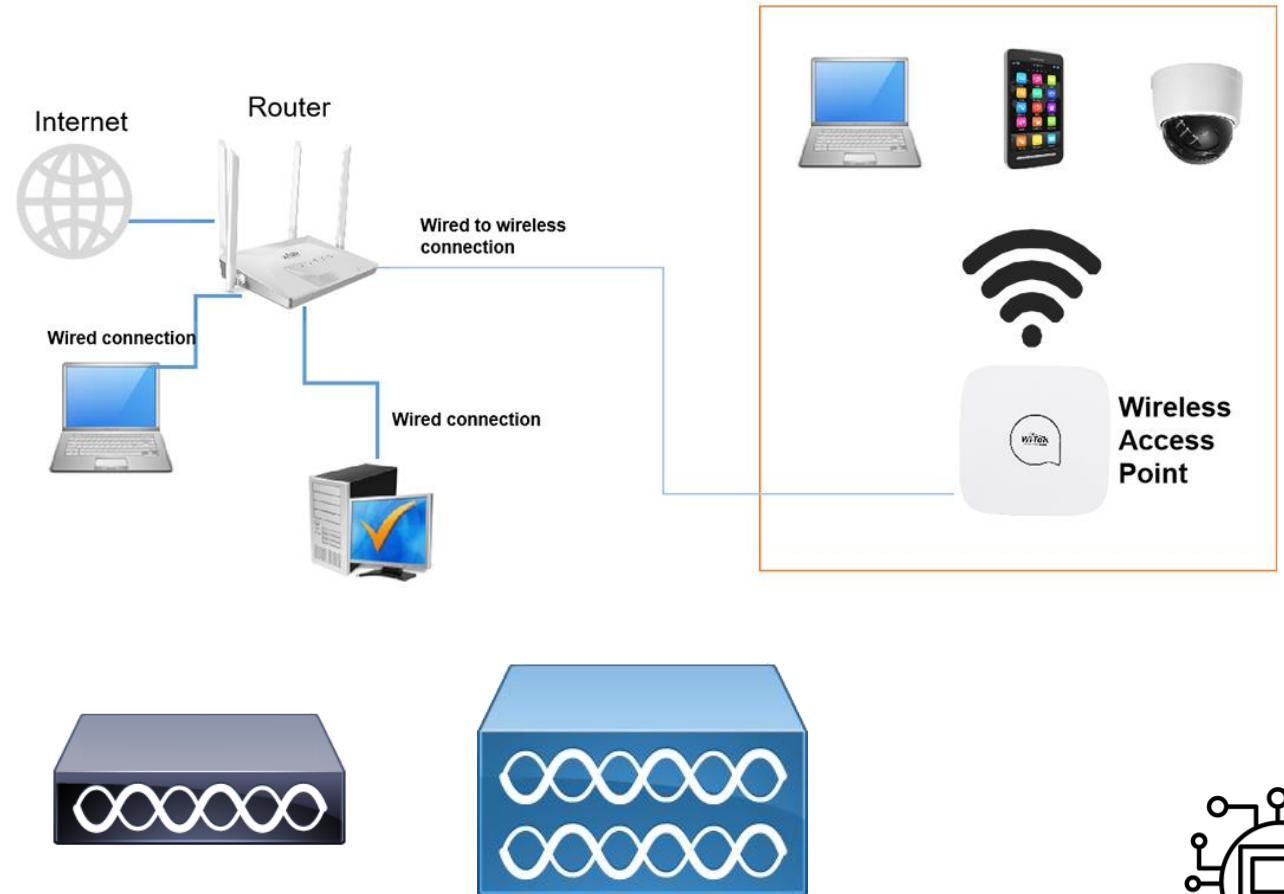
AKA Wireless Access Point, WAP, AP

Allows wireless devices to connect to a wired network

- Acts as a bridge between wired and wireless segments

AP Modes:

- Standalone AP:
 - Independent, one SSID
- Controller-based AP:
 - “Lightweight”, centrally managed, multiple SSIDs
- Cloud-managed AP:
 - Good for multiple sites, remote management
- Mesh AP:
 - One SSID, large area coverage



Wireless LAN Controller (WLC)

Used to manage multiple wireless access points (WAPs)

- Can be hardware, software, or cloud-based
- WAPs communicate with WLC via VPN on LAN

Centralized control of RF, load balancing, security policy, authentication, captive portals, and client roaming

Multiple SSIDs and VLANs on each WAP

Cisco Unified Wireless Network (CUWN):

- Integrated solution of controllers, WAPs, management and security
- Dynamically assigns channels based on client load and background noise



Cisco 9800 WLC



Endpoint

A generic term that refers to any device that connects to, and uses, a network

- Tends to be more associated with a client device as opposed to a server
- Does not include devices that move network traffic such as routers, switches, firewalls, WAPs, IDS/IPS, etc.

Can include:

- Mobile devices
- Laptops
- PCs
- IoT devices
- IP phones and IP cameras
- Printers, TVs



Server

A computer dedicated to providing services to client devices

- Can be physical or virtual

Can provide one or multiple services

Each service “listens” on its own port

Each service waits for clients to make a connection attempt



Power over Ethernet (PoE)

A technology that allows Ethernet cables to carry electrical power along with data to remote devices

Enables network devices like IP cameras, VoIP phones, wireless access points, IoT and other equipment to be powered and connected using a single Ethernet cable

Simplifies installation, reduces the need for additional electrical wiring

Can be built into the switchports or external injector device

Shuts down the port if maximum power value is exceeded



PoE Standards

802.3af (PoE):

- Delivers up to 15.4 watts of DC power over Cat5 cables
- Suitable for devices like IP phones or simple wireless access points

802.3at (PoE+):

- Provides up to 25.5 watts of power
- Allows for higher power devices like PTZ cameras or advanced wireless access points



PoE Standards

802.3bt (PoE++):

- Introduced two new types:
 - Type 3: Delivers up to 60 watts of power.
 - Type 4: Delivers up to 100 watts of
 - Powers more demanding devices such as video conferencing systems or small computers
- Power can be carried on the:
 - data pairs (phantom technique)
 - spare pairs, depending on the mode

1.5 Network Topology

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Module 1

Three-tier

Two-tier

Spine-leaf

WAN

SOHO

On-premises and Cloud

Three-tier Network Topology

Traffic path: Access → Distribution → Core → Distribution → Access

Core Layer:

- The backbone that connects multiple sites or network blocks together
- Provides uninterrupted forwarding service and timely data transfer

Distribution Layer:

- Runs Layer 2 and Layer 3 technologies
- Aggregates access layer traffic
- Enforces policy
- Routes between VLANs (logical network segments)

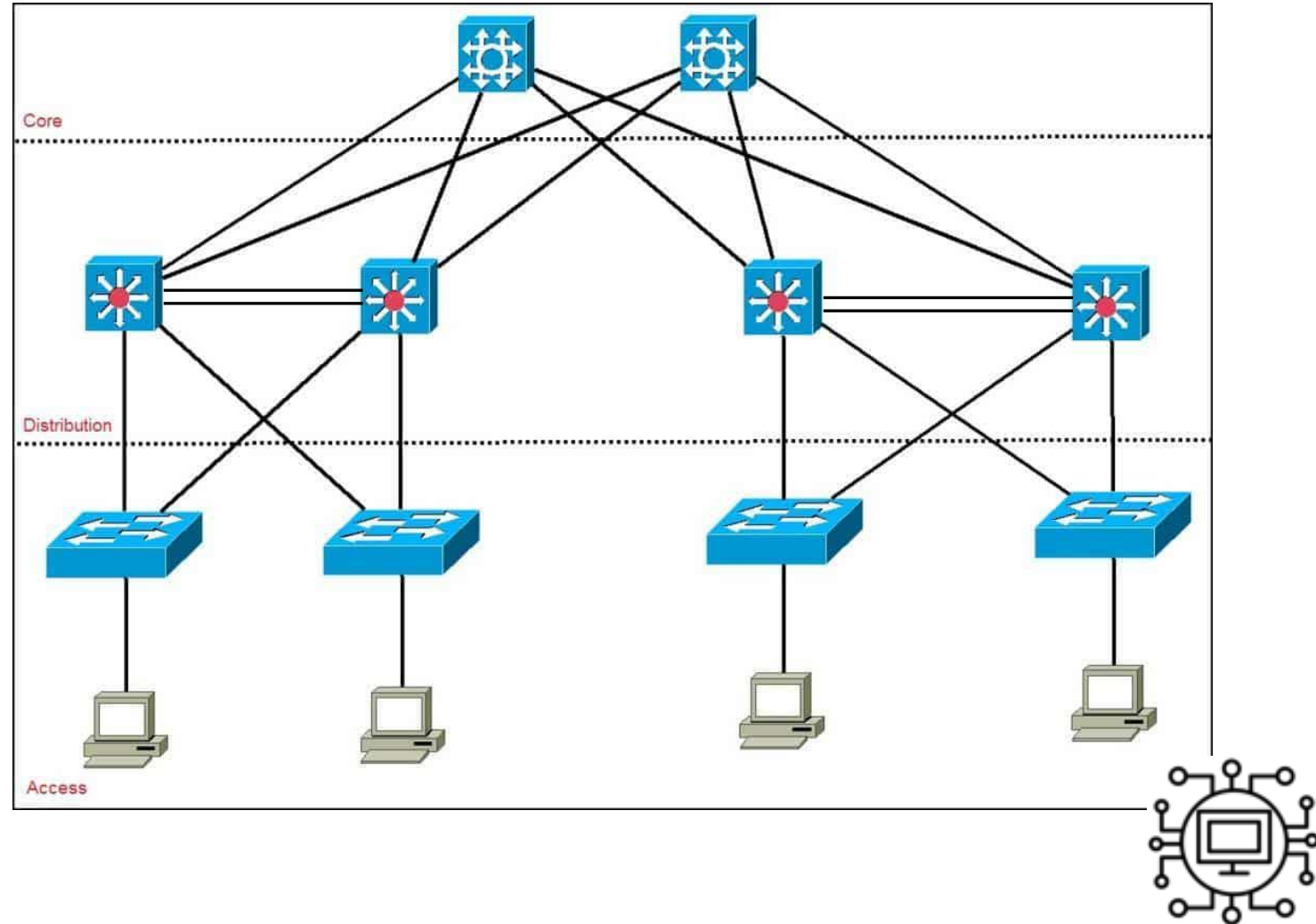
Access Layer:

- Physical connection point for LAN devices
- Different switchports can belong to different VLANs)

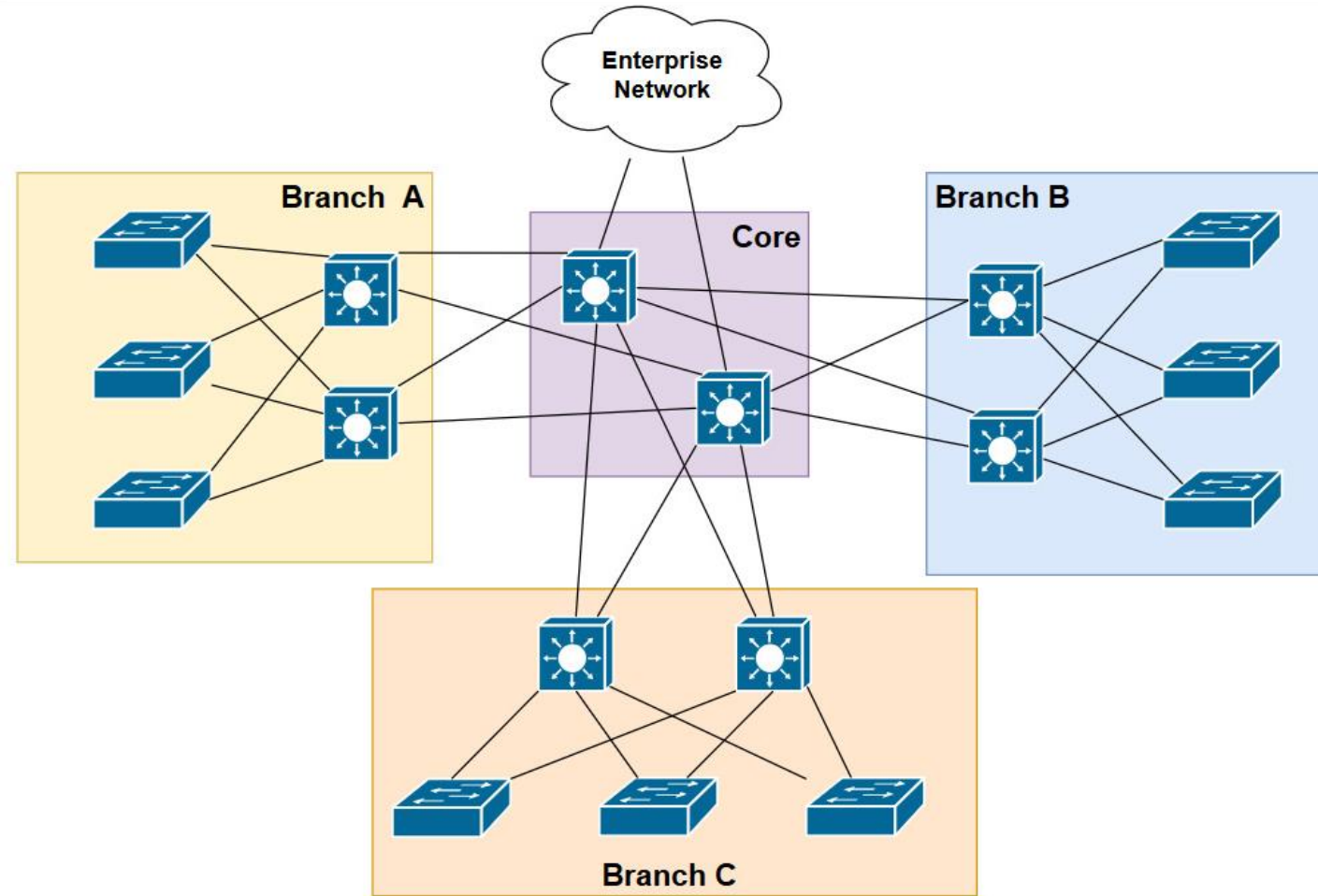


Classic Cisco Three-Tier Model

Tier	Description
Core	• Datacenter backbone • High-speed switching • Connect to campus core or edge router
Distribution	• VLAN Routing • Policy • Redundancy for Access layer • Traffic aggregation
Access	• Connectivity for end devices (clients and servers) • VLAN membership



Three-Tier Example

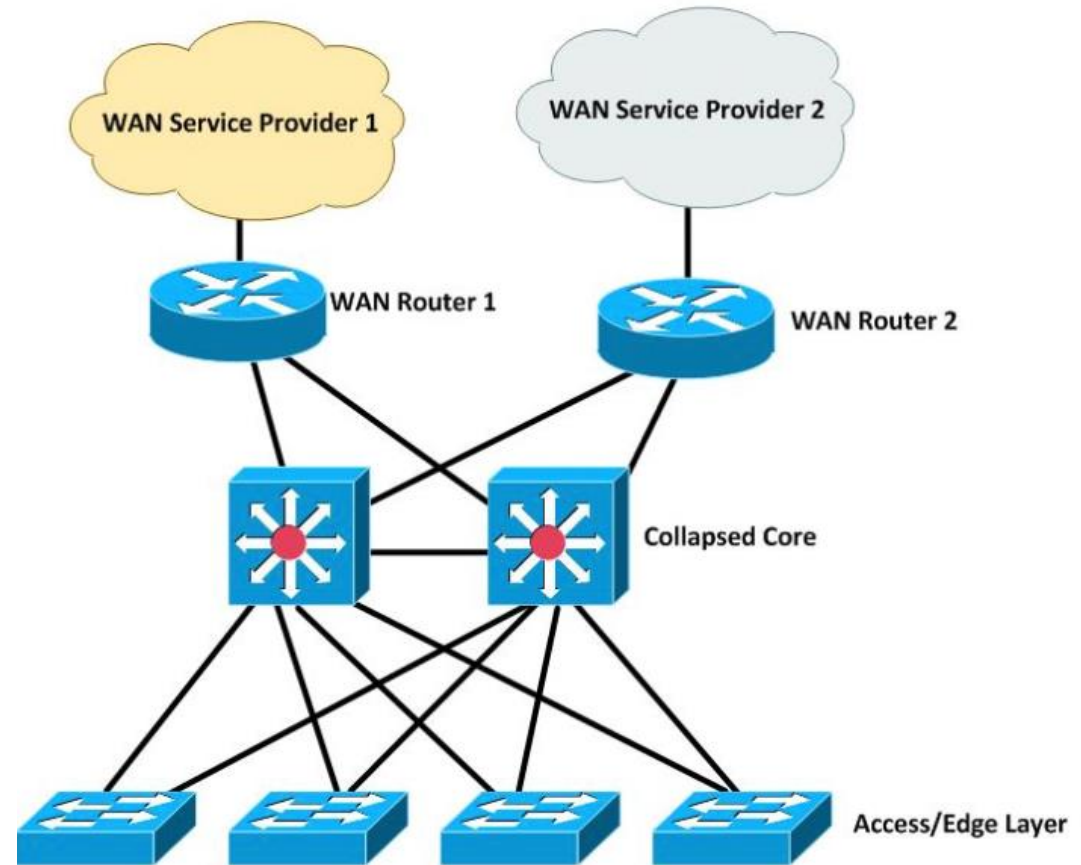


Two-tier Network Topology

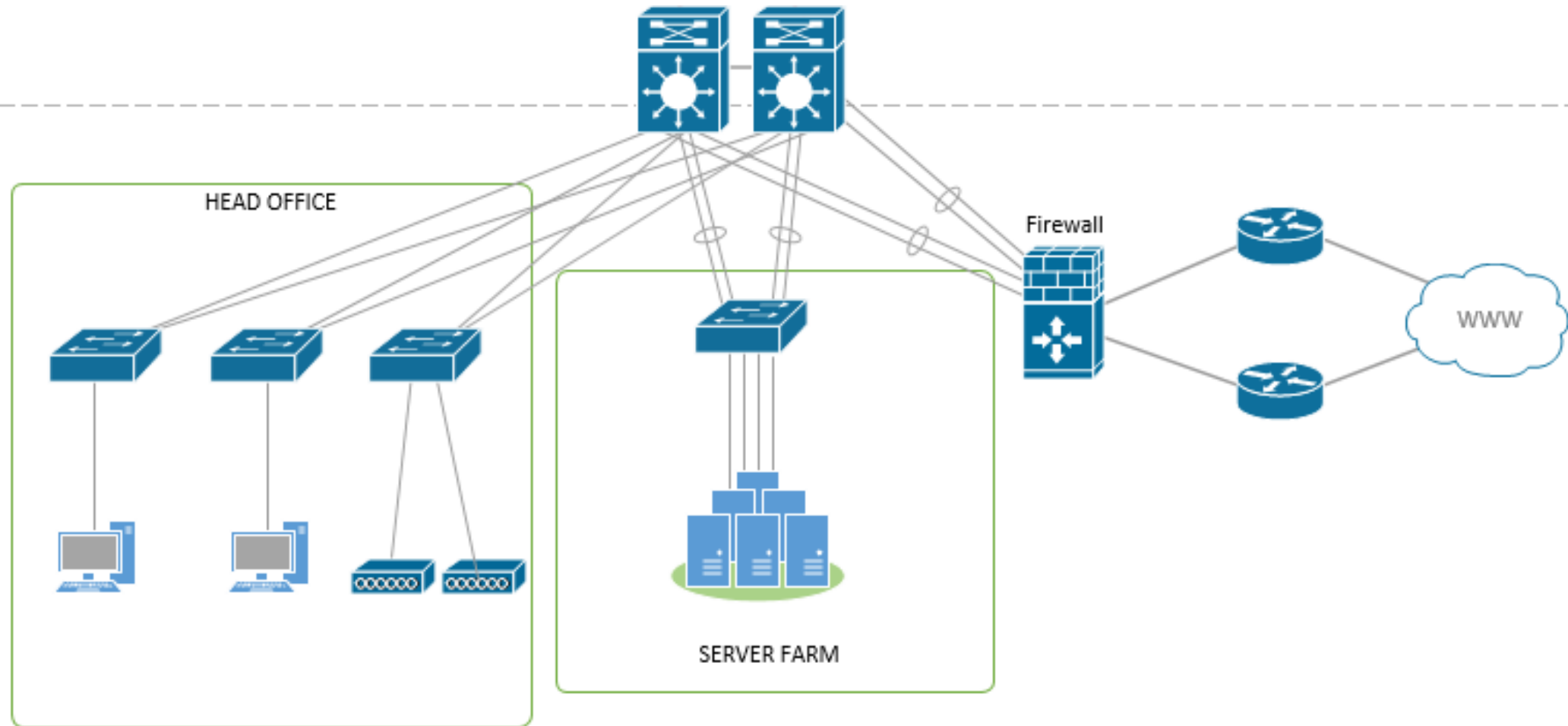
AKA Collapsed-Core

Suitable for small organizations to reduce networking costs.

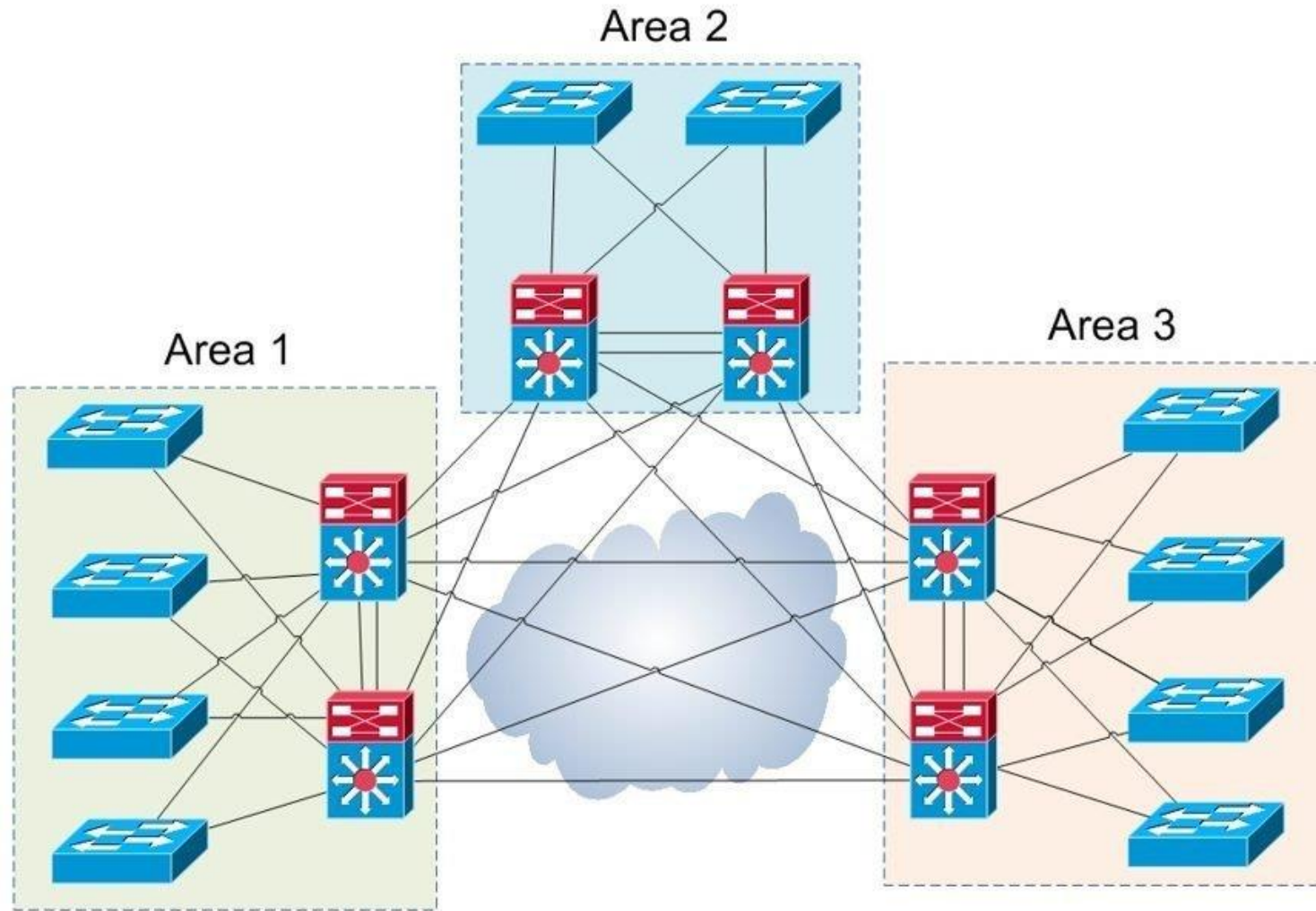
Combines core and distribution layers



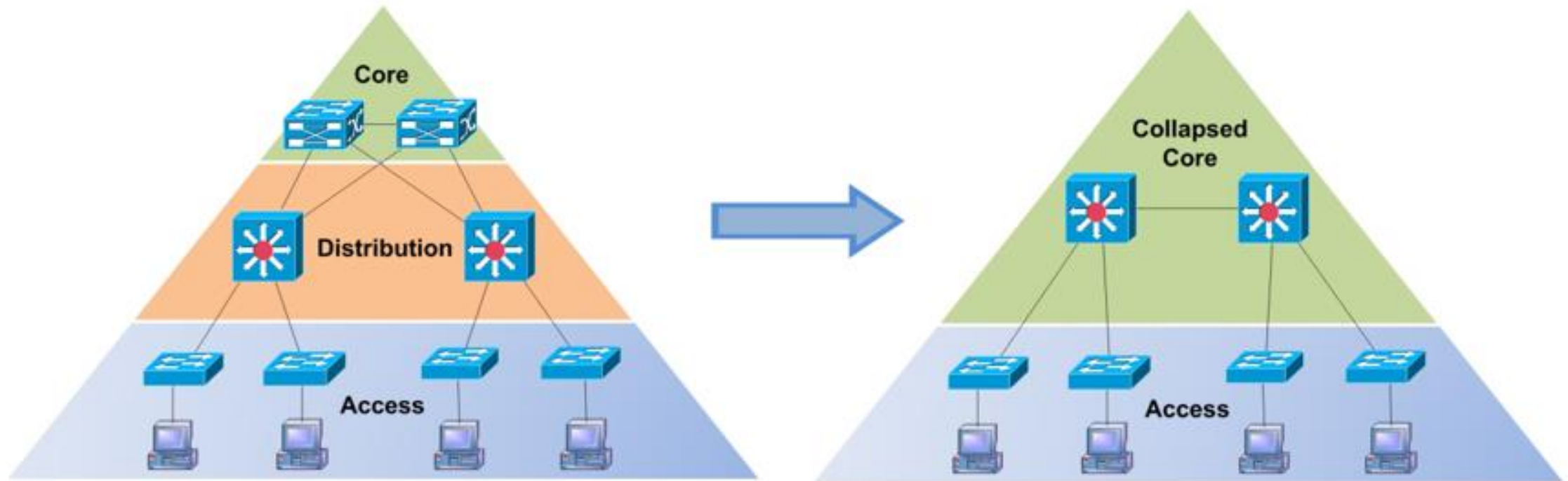
Collapsed Core Example



Collapsed Core with Full-Mesh Distribution Layer



Three-tier vs. Two-tier



Spine-Leaf

AKA Spine-and-Leaf, Leaf-Spine

Two switching layers – spine and leaf

- Each leaf switch connects to every spine switch
- Allows scalability by adding a leaf switch

Leaf Layer:

- Aggregate traffic from servers
- Connects directly to spine (core)

Spine Layer:

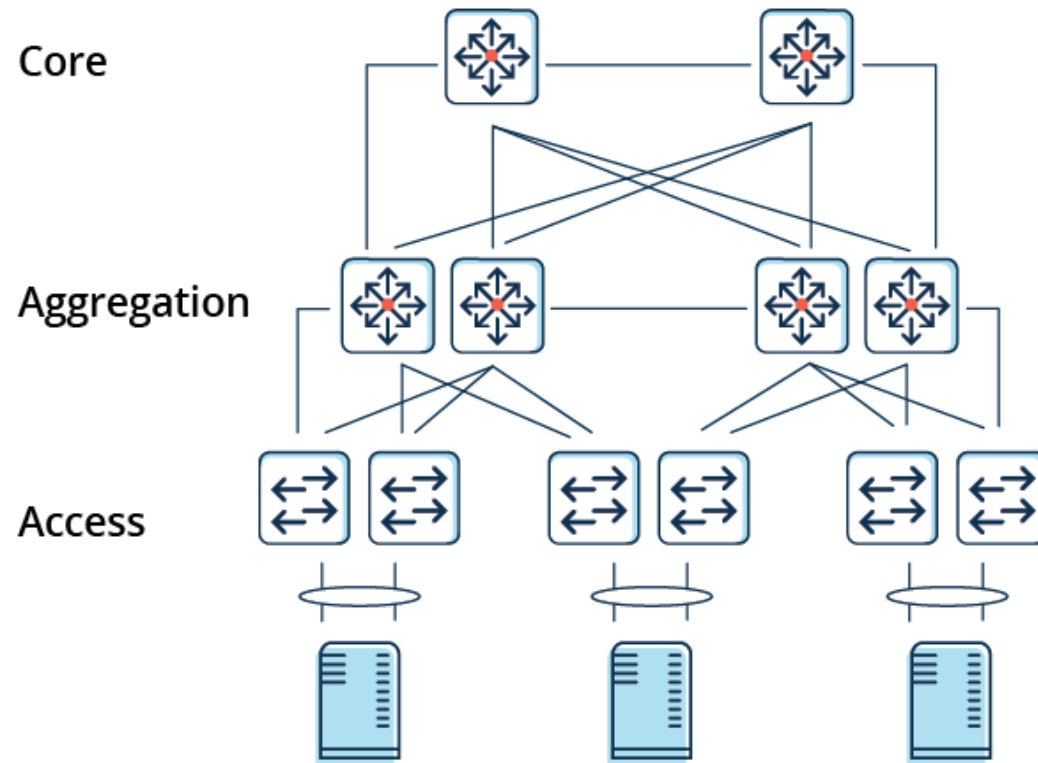
- Spine switches interconnect all leaf switches
- Full mesh topology
- Traffic flow is more East-West than traditional North-South

- No more spanning tree
- Predictable latency
 - Same number of hops between devices
- Load balancing across all available paths
- Physical links must be 25- 40- or 100gb

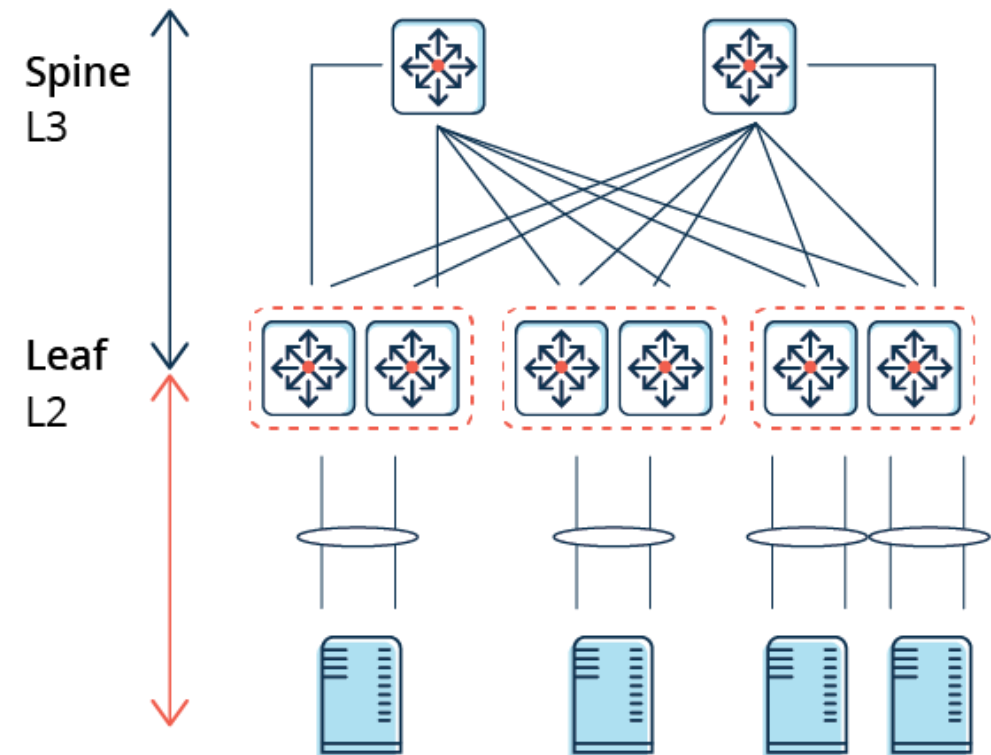


Traditional 3-Tier vs 2-Tier Spine-Leaf

Traditional 3-Tier Architecture



2-Tier Spine-Leaf Architecture



WAN Topology

Point-to-Point

- T3, T1, Fractional T1, Dialup, site-to-site VPN, user-to-site VPN

Hub and spoke (star)

- One central node (hub)
- All other nodes (spokes) connect to the hub
- All communication between spokes is routed through the hub
- Combines simplicity, quality, and availability

Partial mesh

- Redundant star
- Some spokes also have a direct connection to each other

Full mesh:

- All nodes connected to all nodes
- Highest degree of reliability

Dual-homed branches and dynamic routing

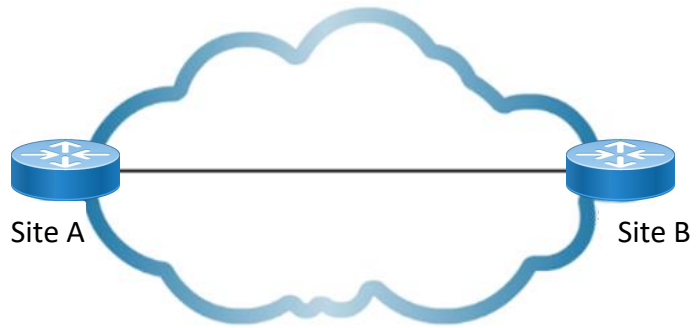
- Hub node has multiple routers
- Spoke nodes connected to both routers in the hub
- Load balance and/or routing protocol chooses preferred link
- Provides extra scalability and reliability

6 to 4 tunneling

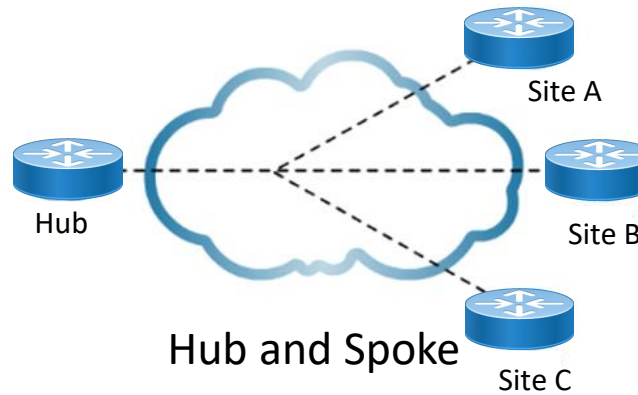
- IPv6 logical topology overlay on top of IPv4 physical topology
- IPv6 traffic routed over IPv4 infrastructure
- Physical and logical topologies can be anything



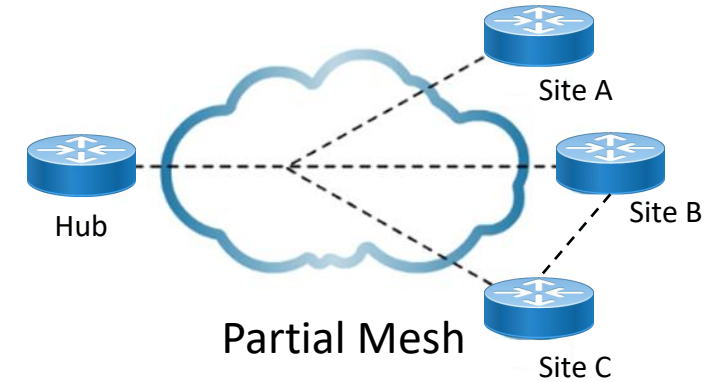
WAN Topology Examples



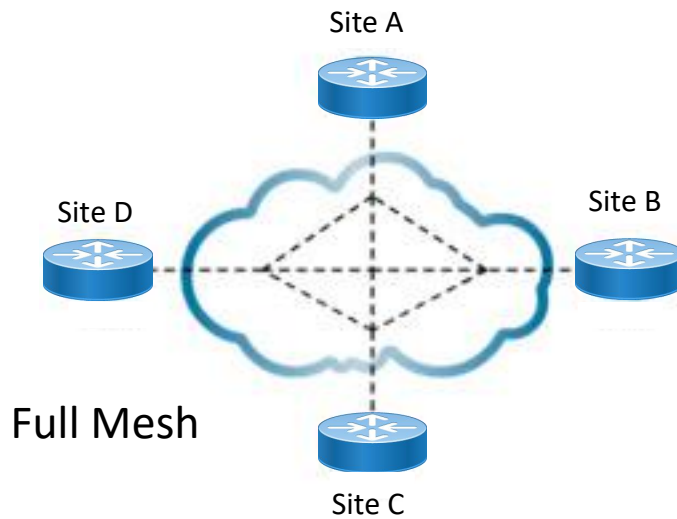
Point-to-Point



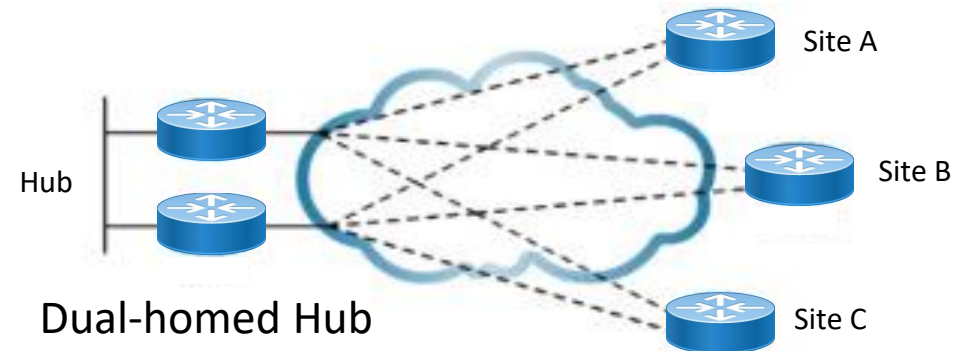
Hub and Spoke



Partial Mesh



Full Mesh



Dual-homed Hub



Small Office/Home Office (SOHO) Topology

Supports multiple users sharing a single broadband connection

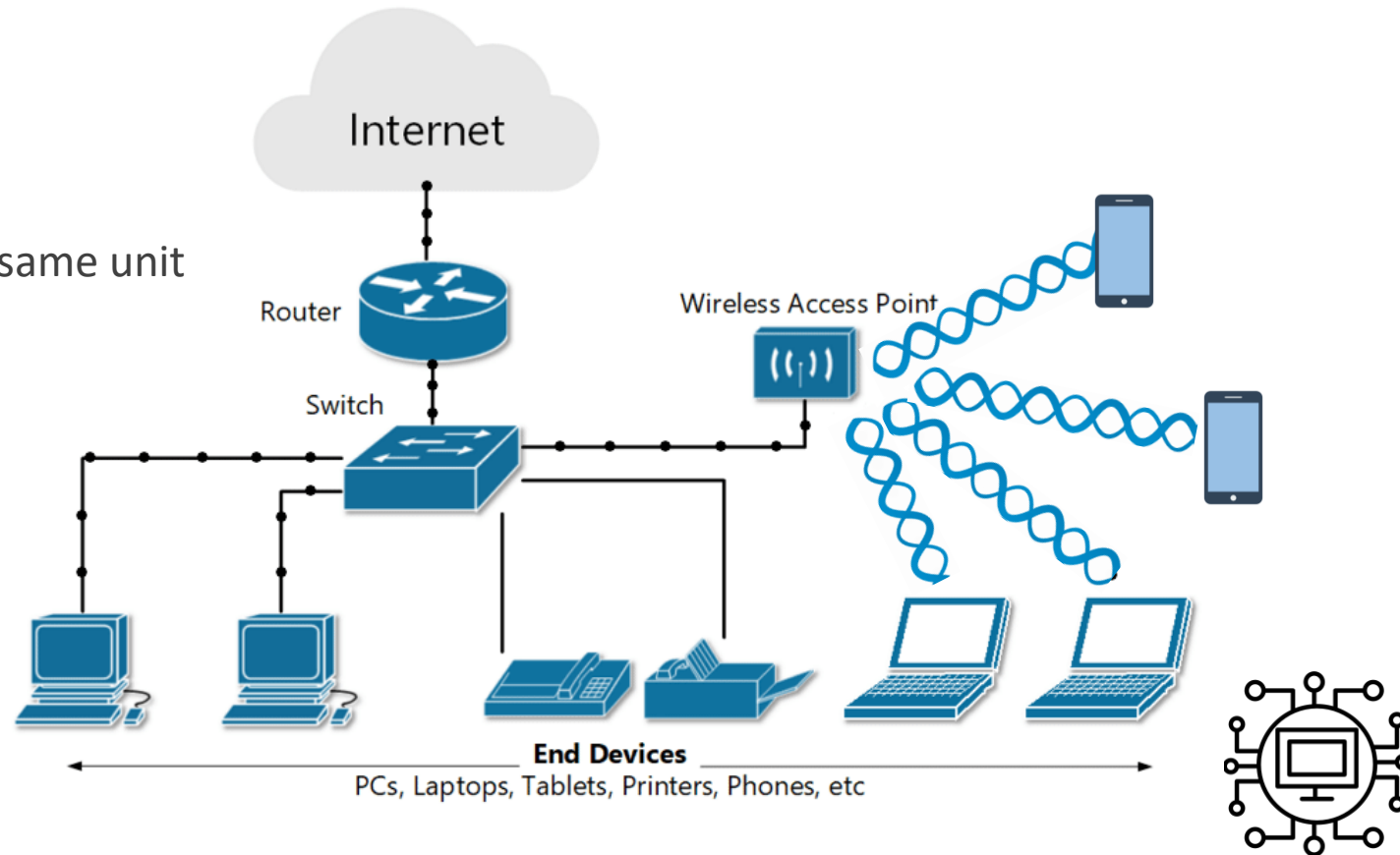
- Network Address Translation

Preferred WAN access technology:

- Broadband cable access or DSL
- Router, WAP and Switch are typically the same unit

DHCP usually provided by the router

Router may or may not have a firewall



On-premises and Cloud Network Topology

Typically public cloud-based

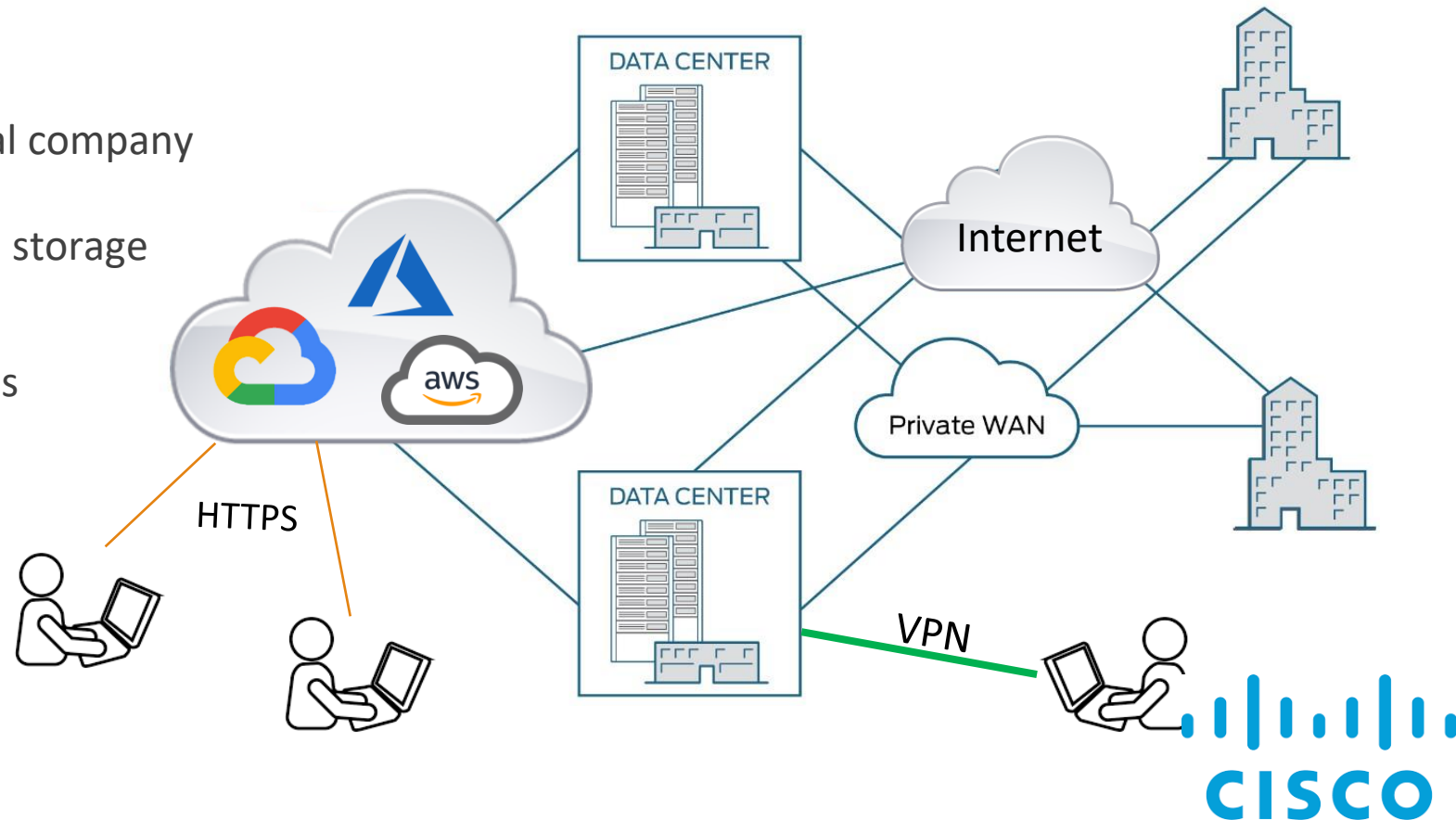
- Cloud is accessed over the Internet

For internal users consuming internal company resources hosted in the cloud

- MS Teams, M365, virtual servers, storage

For external users consuming publicly-available company resources

- Public websites, extranets, video conferences, learning content



1.6 Copper Cable Types

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Module 1

Ethernet Connections

Ethernet over Copper Standards

Twisted Pair

Coaxial

Twinaxial

Network Speed and Storage

Ethernet Connections

Ethernet is transmitted over wired connections

- Can be copper or optical fiber

Point-to-point connections

- One interface to one interface
- E.g. computer to switch, router to switch, firewall to router, switch to switch, IP phone to IP PBX, IP camera to Network Video Recorder (NVR), etc.
 - Also: WAN link, crossover cable between two PCs, two switches, two routers, etc.

Shared media connections

- A common physical or logical connection in which all nodes “hear” all traffic from all other nodes
- Can be:
 - Multiple nodes tapped into the same conducting cable (typically coaxial cable)
 - Multiple nodes plugged into a hub
 - Multiple nodes on the same radio (Wi-Fi) channel



Ethernet over Copper Standards

Copper Standard	Name	Max Distance	Description
10Base-T	Standard 10 megabit Ethernet	100m	IEEE 802.3 CAT3/4/5 UTP Uses 2 pairs
100Base-TX	100 megabit Fast Ethernet over twisted pair	100m	IEEE 802.3u CAT5 UTP/STP Uses 2 pairs
1000Base-T	Gigabit Ethernet	100m	802.3z, 802.3ab CAT 5e UTP Uses all 4 pairs
10GBase-T	10 Gigabit Ethernet	100m	802.3an CAT 6 or above
40GBase-T	40 Gigabit Ethernet	30m	802.3bq CAT 7a/8

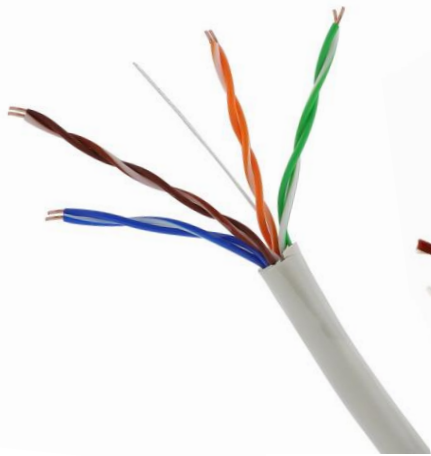


Twisted Pair Cable Categories

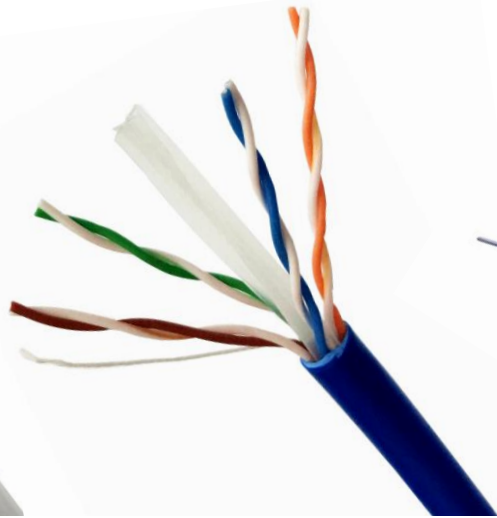
Category	Data Rate	Application
CAT 1	Up to 1 Mb/s	Old telephone system
CAT 2	4 Mb/s	Token Ring
CAT 3	10 Mb/s	Ethernet
CAT 4	16 Mb/s	Token Ring
CAT 5	100 Mb/s	Fast Ethernet
CAT 5e	1000 Mb/s	Gigabit Ethernet
CAT 6	Up to 10 Gb/s	Gigabit Ethernet 10gb @ 55 meters
CAT 6a	Up to 10 Gb/s	1gb / 10gb Ethernet Especially for IP cameras & WAPS at longer distance
CAT 7	Up to 10 Gb/s	Outdoor or industrial Ethernet environments ScTP up to 100 meters
CAT 8	Up to 40 Gb/s	Short distance – 5 – 30 meters



Comparing Twisted-Pair Cable Categories



CAT 5e



CAT6



CAT6a



CAT7



CAT8

TIA/EIA 568B = Orange pair - transmit; Green pair - receive



Coaxial Cable (Coax)

The standard for *original* Ethernet networks

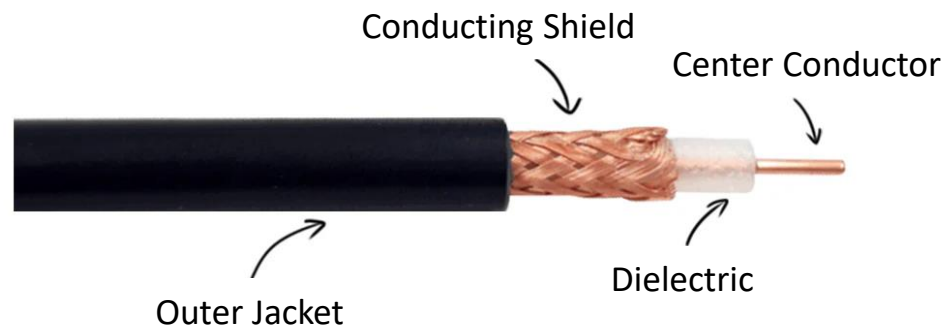
Used today in cable TV/cable modems, radio

Coaxial cables consist of inner copper wire of various thickness

- Covered by insulation and shielding

The stiffness made coaxial cable difficult to install and maintain

Different impedances make certain cable types incompatible



Coaxial F connectors



Coaxial BNC connectors



Twinaxial (Twinax)

A variation on coax

Low cost alternative to fiber optic

Good choice in datacenters

High speed - 1gb/s, 10gb/s

Short distance – 5 meters max

SFP+ connectors

Twin conductors work together

- Both transmit half duplex
- Rising edge of bit – A has a higher voltage
- Falling edge of same bit – B has a higher voltage



Proper Way to Indicate Speed and Storage

Network speed

Measured in bits per second

- With a small “b”
- Examples:
 - 56 kb/s – 56 kilobits per second
 - 128 Kbps – 128 kilobits per second
 - 11 Mbs – 11 megabits per second
 - 100 Mb/s – 100 megabits per second
 - 10 Gb/s – 10 gigabits per second

Storage (file and drive size)

Measured in bytes

- With a capital “B”
- Examples:
 - 64 KB – sixty four kilobytes
 - 45 MB – forty five megabytes
 - 1 GB – one gigabyte
 - 2 TB – two terabytes

1.7 Fiber Optic Cable Types

CCNA 200-301 v1.1

Module 1

Ethernet over Fiber Optic Standards

Fiber Optic Cable

Multimode Fiber

Single Mode Fiber

Fiber Full Duplex

Ethernet over Fiber Optic Standards

Fiber Standard	Max Distance	Description
100BASE-FX	10 k	Fast Ethernet (100 mb)
100Base-SX	300 m	Lower cost Fast Ethernet Backward compatible with 10base-FL
1000Base-SX	550 m	Gigabit Ethernet
1000Base-LX	2 km	Gigabit Ethernet long-wavelength (LX) transmission
10GBase-SR	400 m	10 Gigabit Ethernet
10GBase-LR	10 k	10 Gigabit Ethernet Long-range (LR) transmission
40GBase-LR	10 k	40 Gigabit Ethernet LR4 multiplexes 4 x 10G links for a total of 40 gb/s

Note: All of the above require 2 strands: 1 Tx and 1 Rx
Cables may require repeaters to achieve max distance



Fiber Optic Cable

AKA Optical-Fiber cabling

Conductor is a strand of glass or plastic

- Carries pulses of light instead of electrical signals
- Light is fast and is not subject to electrical interference

Common in LAN backbones and trunk links between switches

- You can run fiber-optic cable over very long distances without having to boost or clean the signal

More costly than copper or Twinax

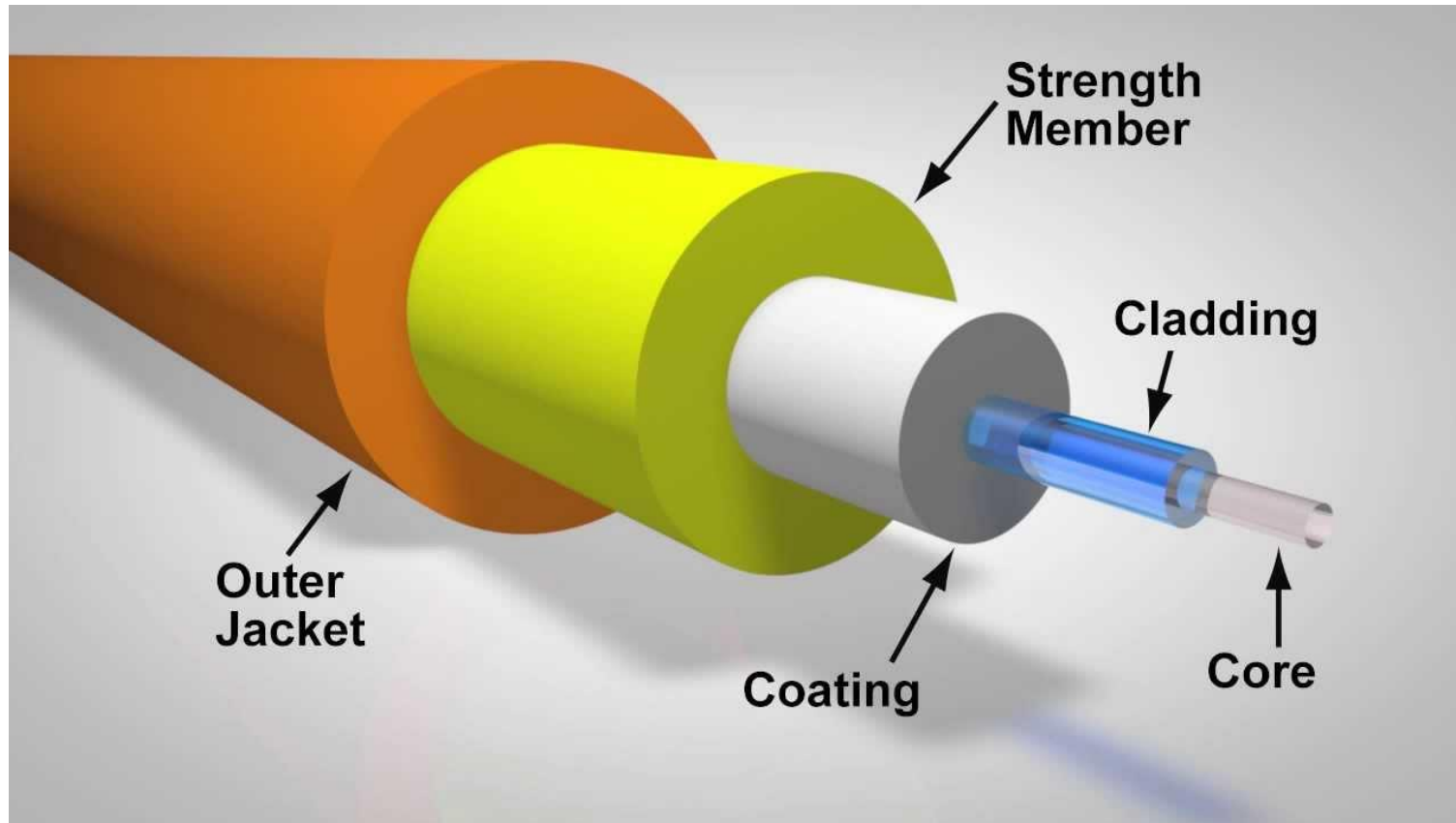
Required by 10GBase-SR and 10GBaseLR

You need two cables or two beams of light for a two-way transmission

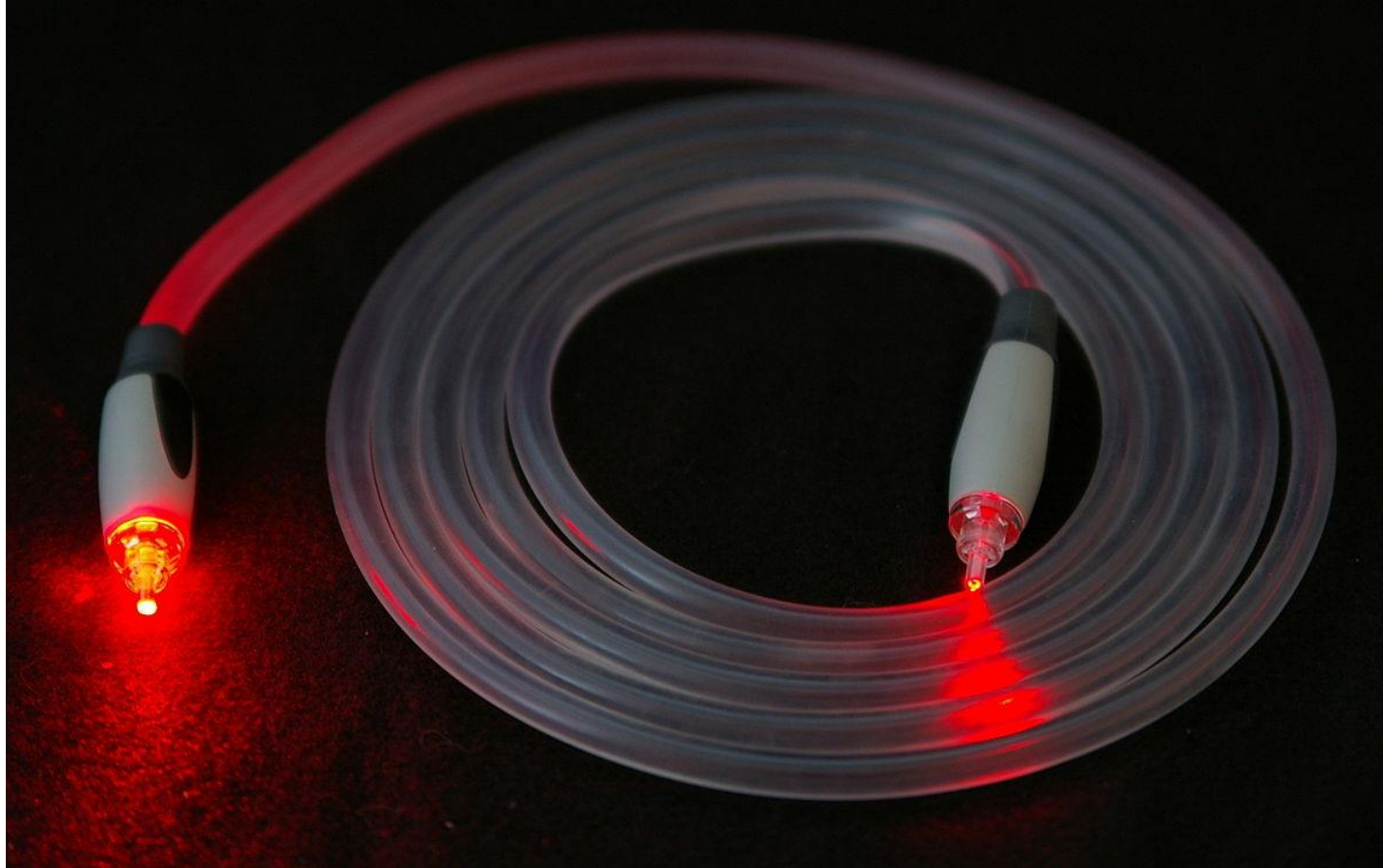
- One for transmitting (Tx)
- One for receiving (Rx)



Fiber Optic Cable Construction



Fiber Optic Light Transmission Example



Multimode Fiber Optic Cable

50 micron core / 125 micron cladding

- Wider diameter allows multiple light beams to travel on the cable
- Easier to terminate
- More possible signal distortion at the end

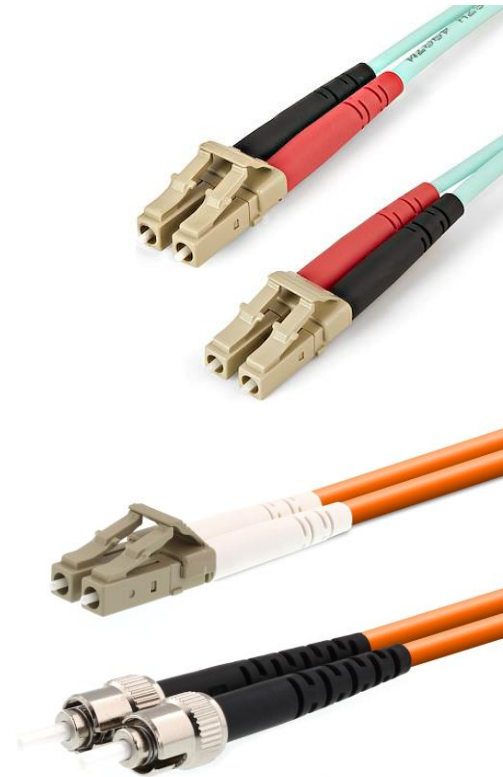
2 km max distance

- **2 km** (100BASE-FX), 1 Gbit/s up to **1000 m**, and 10 Gbit/s up to **550 m**

850 nm and 1300 nm wavelength

Used in LANs

- Typically trunk links between switches



Multimode OM Cable Designations

Type	Core /Cladding (μm)	Fast Ethernet 100 mb	Gigabit GbE	10 Gigabit 10GbE	40 Gigabit 40GbE	100 Gigabit 100GbE	40G SWDM4	100G SWDM4
OM1	62.5/125	2km	275m	33m	-	-	-	-
OM2	50/125	2km	550m	82m	-	-	-	-
OM3	50/125	2km	800m	300m	100m	100m	240m	75m
OM4	50/125	2km	1100m	400m	150m	150m	350m	100m
OM5	50/125	2km	1100m	400m	150m	150m	440m	150m

Colors indicate cable jacket color

Note: Cable colors are different for military applications



Single Mode Fiber Optic Cable

Very narrow core

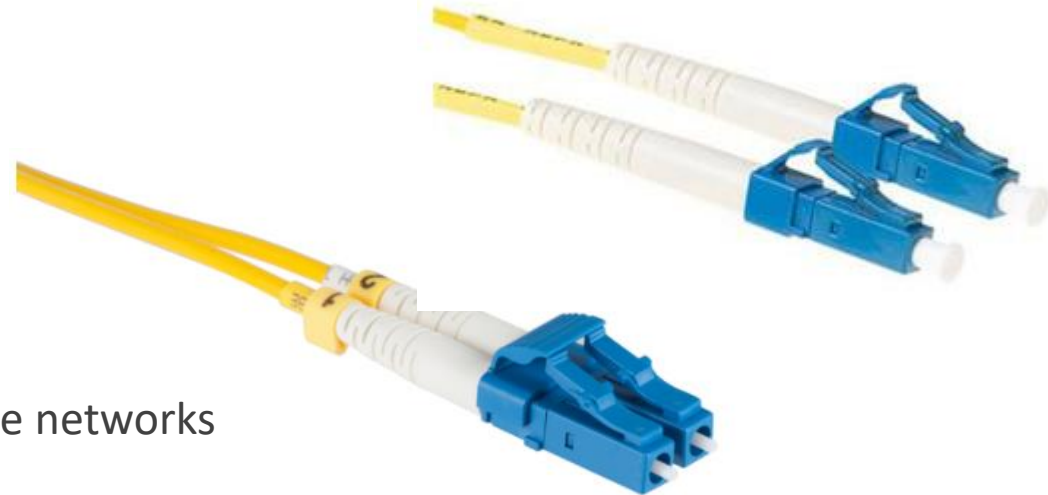
Single beam of light travels through a single path

8 – 10 micron core / 125 micron cladding (designated as 9/125)

Laser light source

Used in:

- LAN backbones
- Metropolitan area networks
- Long-haul and high-data-rate metro networks
- High-speed, high-capacity, unrepeated submarine networks



Single Mode Fiber Optic Cable Designations

OS1 cable

- Indoor
- 10 km max distance
- 1.0 db/km attenuation
- Low price

OS2 cable

- Outdoor
- 200 km max distance
- 0.4 db/km attenuation
- High price

Basic specs are the same

- Difference is cable quality

SMF patch cables are always yellow in color

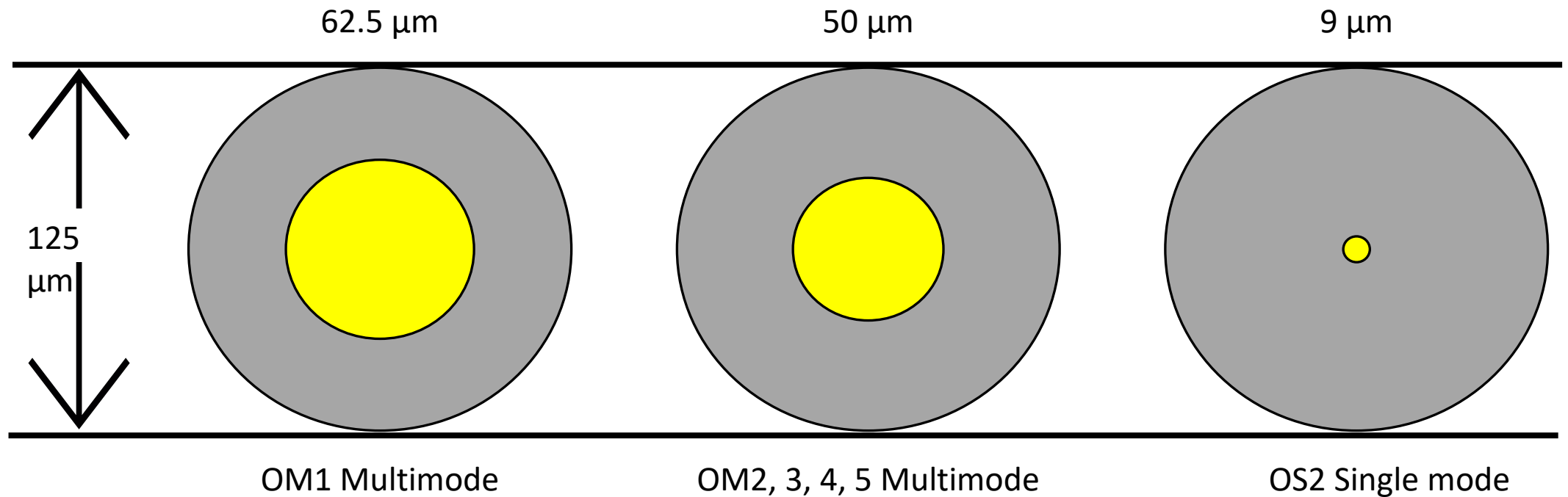


Single Mode Fiber Characteristics

	TYPE	APPLICATION	DISTANCE	WAVELENGTH
OS1	Gigabit	1000BASE-LX	5 km	1310 nm
	10 Gigabit	10GBASE-LX4	10 km	1310 nm
	10 Gigabit	10GBASE-E	40 km	1550 nm
	40 Gigabit	40GBASE-LR4	10 km	1310 nm
OS2	40 Gigabit	40GBASE-FR	2 km	1310 nm
	100 Gigabit	100GBASE-LR4	10 km 40 km	1310 nm 1550 nm



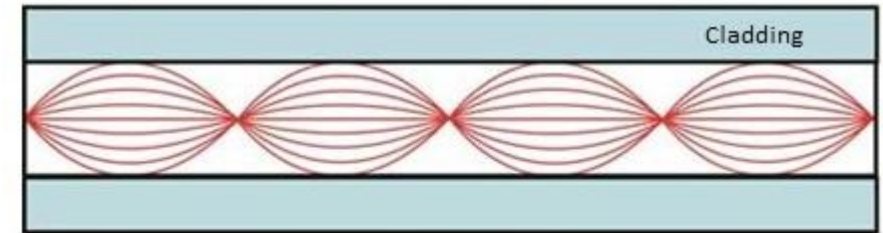
Comparing Fiber Optic Cores



How Light Travels Down a Fiber Optic Cable

Multimode:

- Light travels through a large core in many rays called modes (multiple modes)
- Due to refraction, the rays are reflected from the cladding surface back into the core as they move through the fiber
- More reflection = faster signal degradation = shorter practical distances



Multimode, Graded Index

Single Mode:

- Light travels through a much smaller core
- This forces the light to travel in one ray or mode (a single mode)
- Less reflection = slower signal degradation = longer distances



Singlemode

thefoa.org

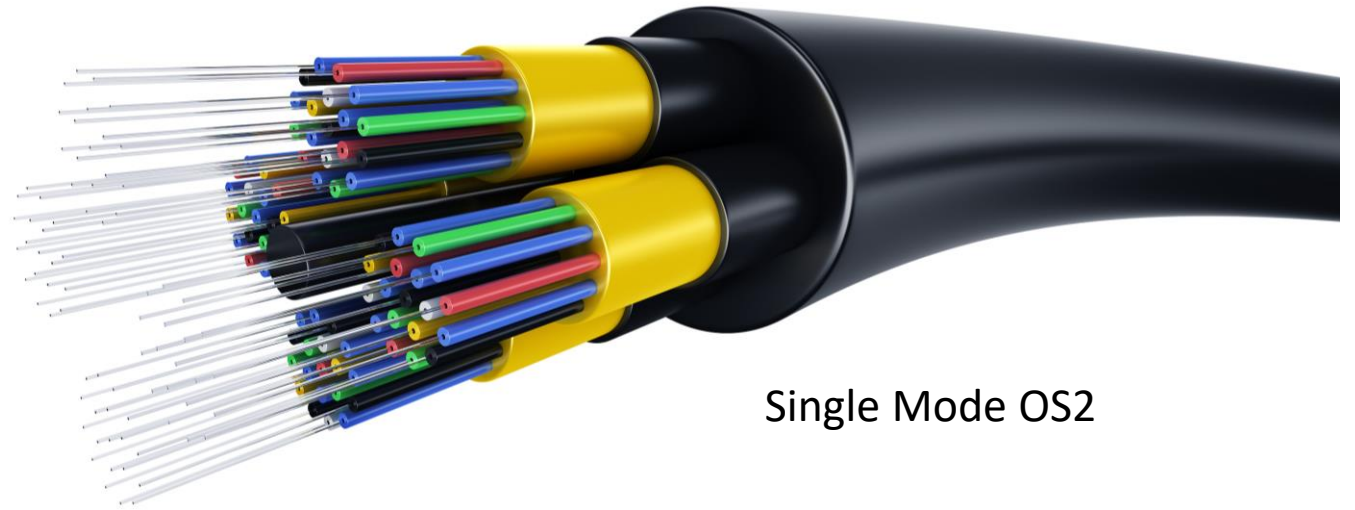


Fiber Optic Backbone Cable

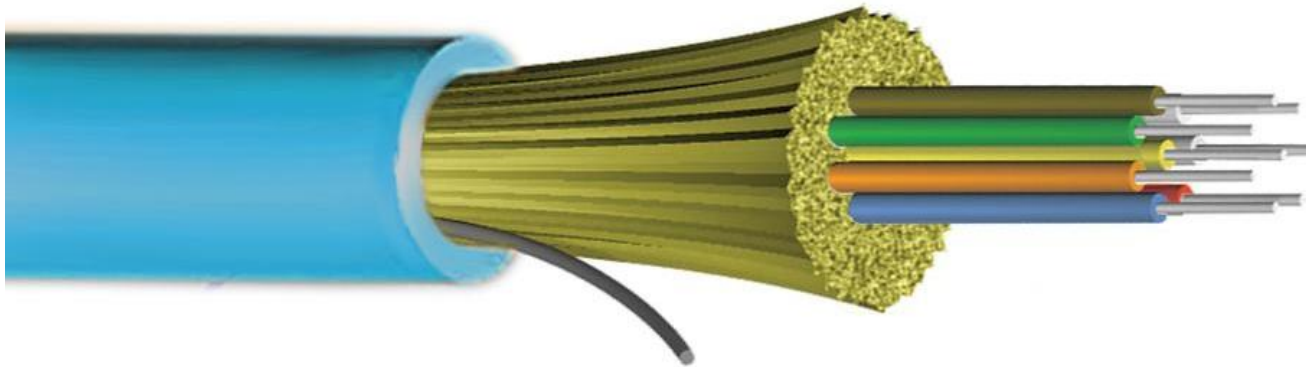
Can be SMF or MMF

2 - 48 strands

Indoor or outdoor



Single Mode OS2



Multimode OM3



Can You Interchange Single- and Multimode Fiber?

Not a good idea!

Most devices don't support it

Cable designed for specific wavelengths

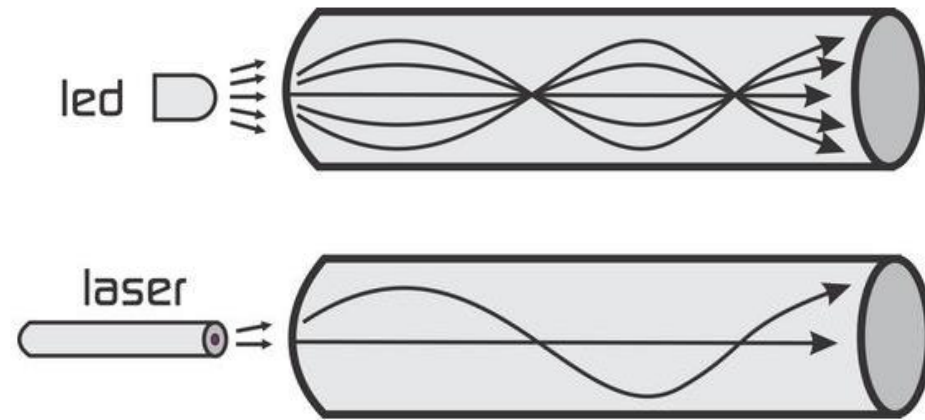
Will degrade signal and reduce distance

SMF signal might overdrive MMF receiver

- Could be too powerful

From end to end, make sure all cable modes, types, and transceivers match!

This includes backbone cable, patch panels, and patch cables!



Wavelength	
MMF	SMF
850 nm	n/a
1300 nm	1310 nm
n/a	1550 nm (1490 – 1625)



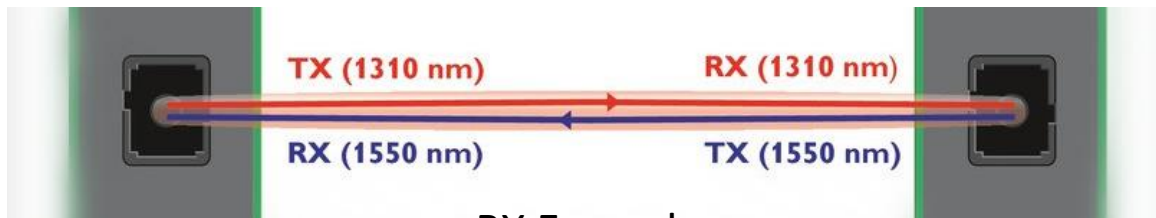
Fiber Full Duplex

Dual-connector SFP

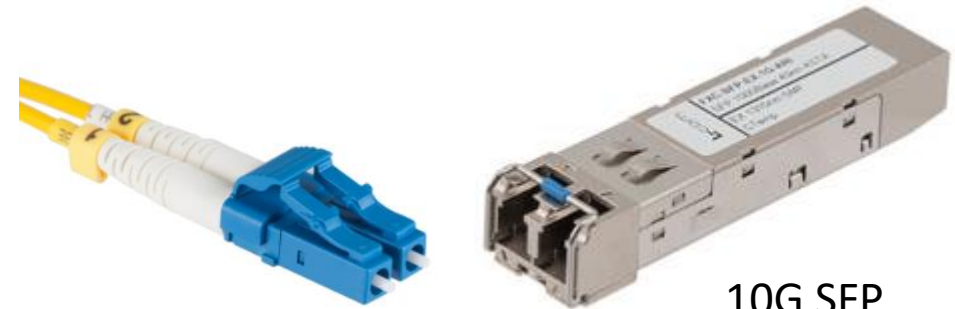
- Each side transmits on its own cable

BX (bi-directional) SFP transceiver

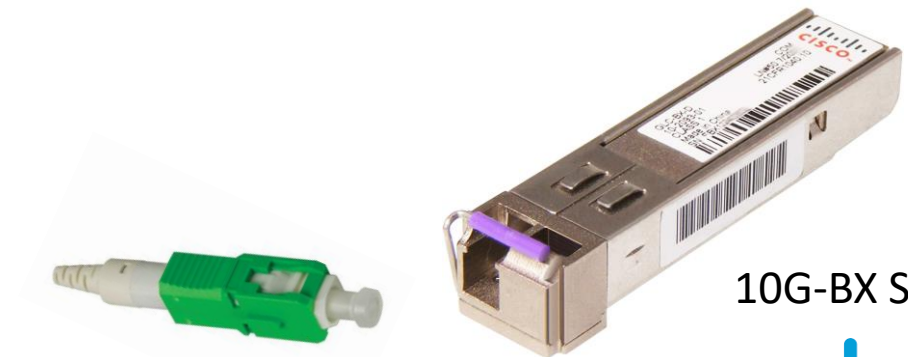
- Transmits/receives on same SMF cable
- One side transmits at 1310 nm (228.849 THz)
- The other side transmits at 1550 nm (193.1 THz)



BX Example



10G SFP



10G-BX SFP

1.8 Virtualization Fundamentals

CCNA 200-301 v1.1

Module 1

Virtual Machines

Server Virtualization

Containers

VRF

Virtual Machines

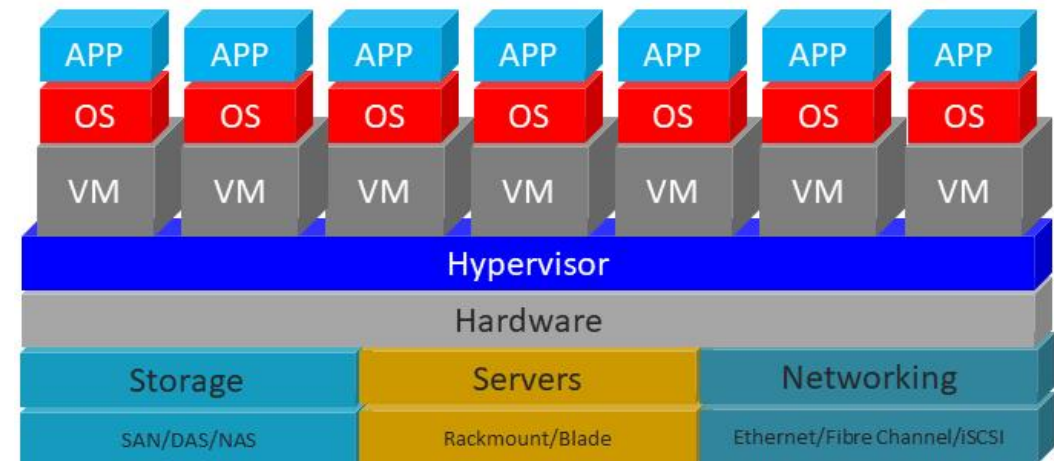
Requires a hypervisor for hardware and virtual machine communication

- Hypervisor virtualizes physical components like CPU, memory, and storage
- Guest OS thinks it's running on its own hardware
- Has its own network interface(s), MAC address(es), IP address(es)
- Appears on the network as a “regular” computer

Saves datacenter floor space

Reduces power consumption

CPU, RAM dynamically allocated as needed



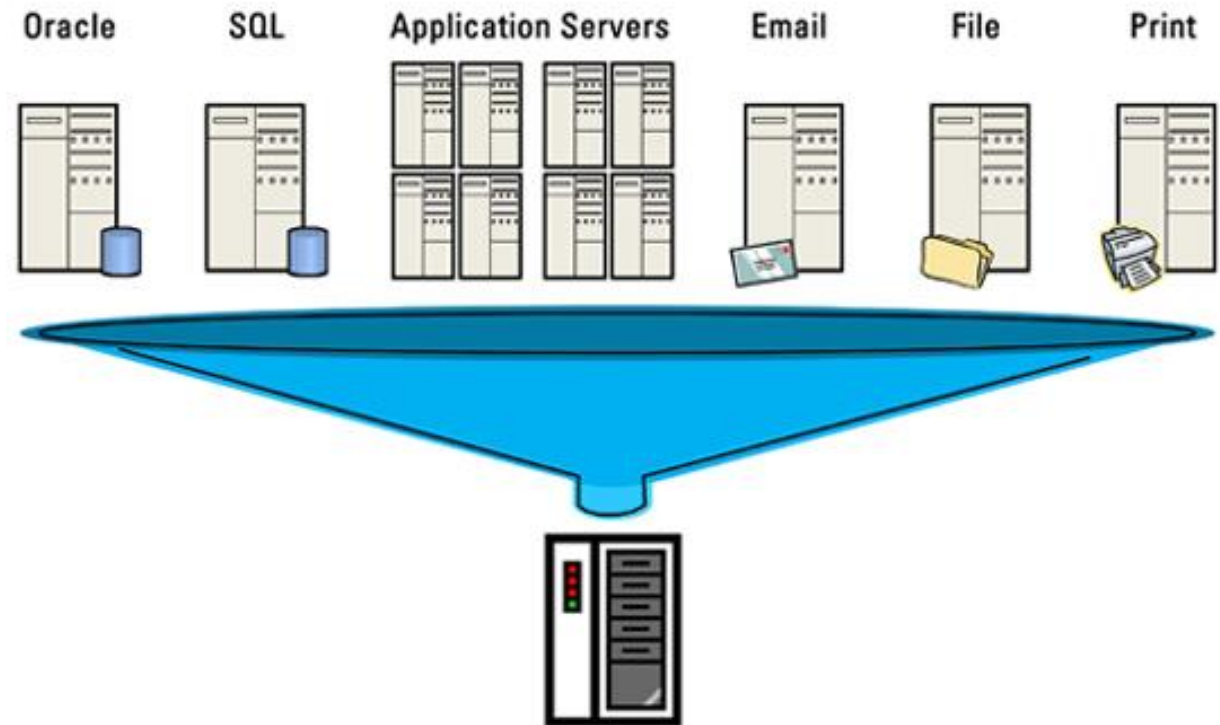
Server Virtualization

A virtual server behaves like a physical server

- Has its own MAC/IP addresses
- Runs applications and services
- Sends and receives data
- Can service multiple clients

Easy to provide redundancy

Dramatically reduces space and power consumption



Containers

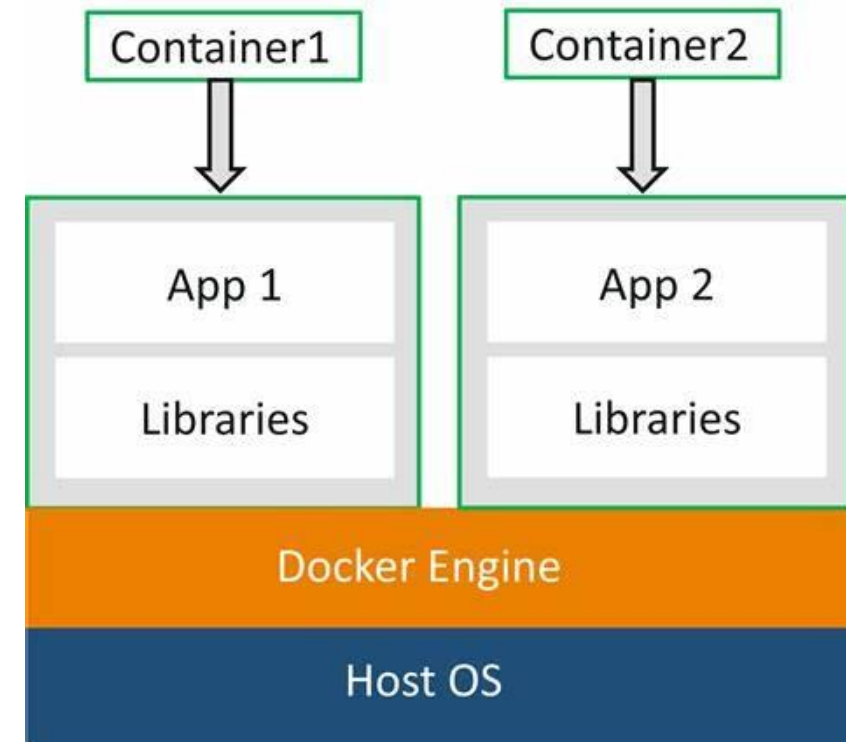
A “virtual machine” for one application

Contains the app and just enough files to run it

- Lightweight – no dedicated OS
- Shares the host OS kernel

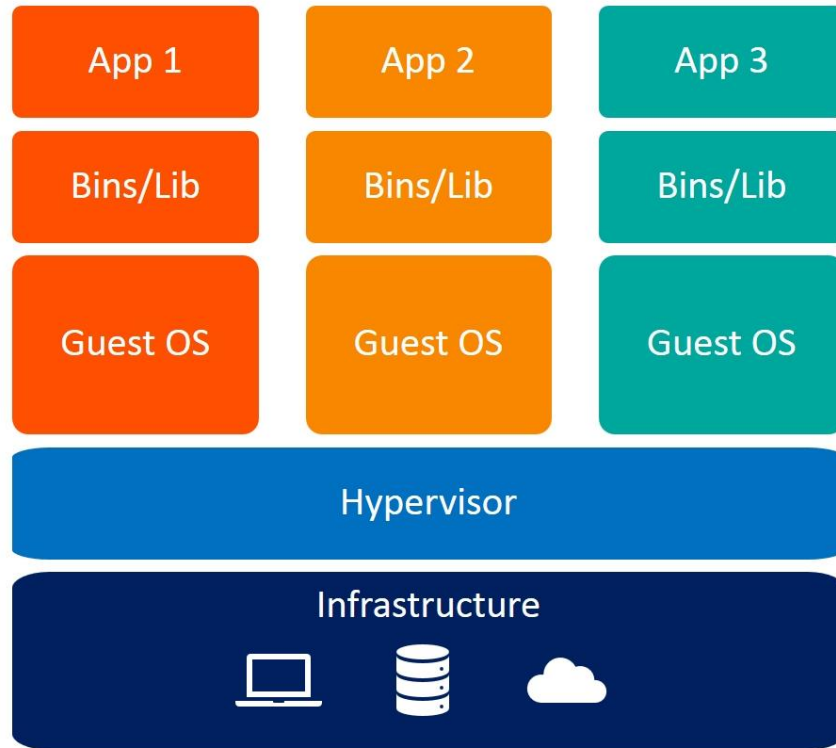
Example container products include:

- Docker
- OpenVZ
- LXC
- ZeroVM

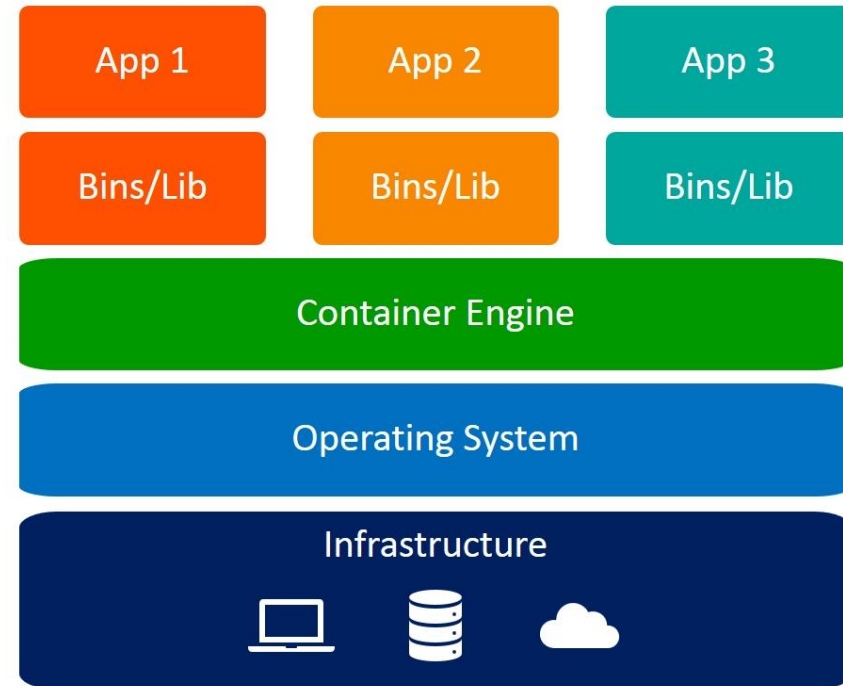


VMs vs. Containers

Virtual Machines



Containers



Virtual Routing and Forwarding (VRF)

A technology that allows multiple instances of a routing table to co-exist within the same router at the same time

VRFs are the TCP/IP layer 3 equivalent of a VLAN

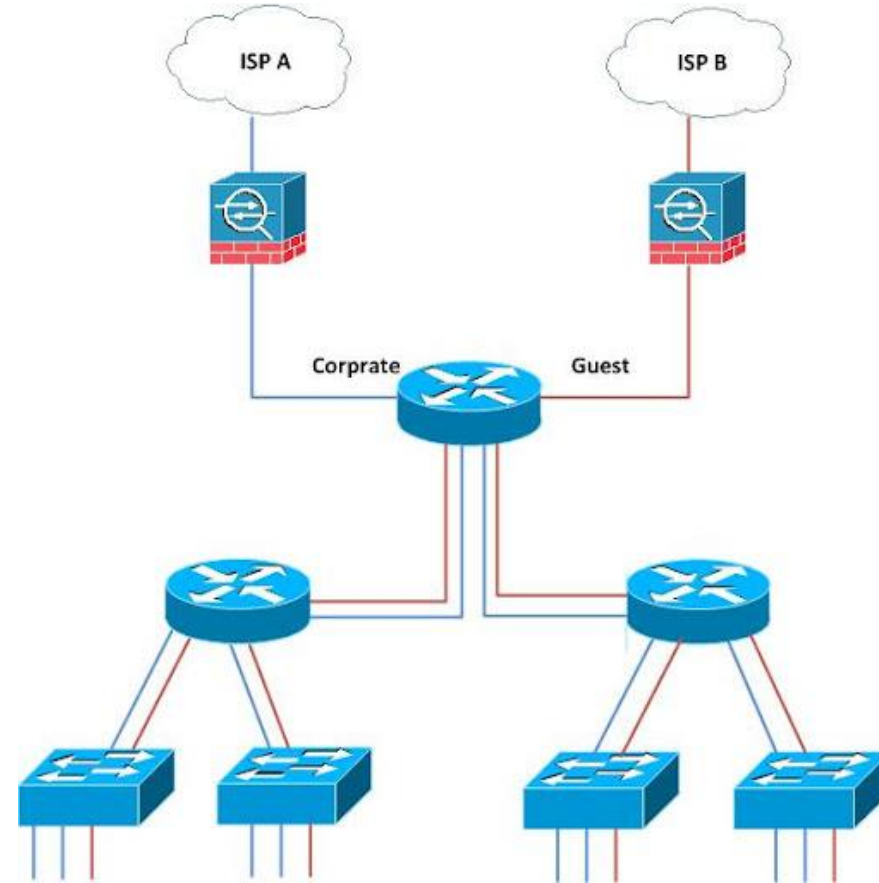
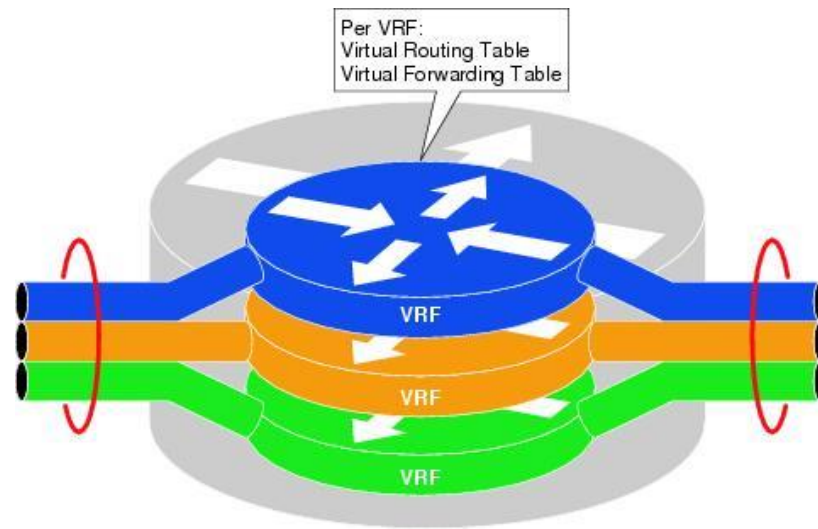
- Provide a fully isolated network path
- One or more logical or physical interfaces may have a VRF and these VRFs do not share routes
- Packets are only forwarded between interfaces on the same VRF
- The same or overlapping IP addresses can be used without conflicting with each other
- Network paths can be segmented without requiring multiple routers

Implemented over a common backbone using VPNs

- Routers encapsulate traffic and mark it to identify the VRF instance
- Common in service providers MPLS networks to isolate different customers
- Can also be used in a corporate environment as VRF-lite



VRF Examples



Review

1.9 Review

Network Fundamentals

CCNA 200-301 v1.1

Module 1

Review

A network consists of two or more devices that communicate across a wired or wireless connection

To communicate, the devices must have corresponding applications, common protocol(s), network interfaces, and a transmission medium that connects them

An IP address represents a node on the entire network

A MAC address represents a node on a single network segment

A port represents a process on the network



Review (cont'd)

Server processes listen on well-known ports

Client processes are temporarily assigned a port by their operating system

A protocol is a common language used to communicate on a network

Protocols exist at all levels of networking

A client is the node or app that initiates a connection

A server is the node or app that waits for a connection attempt by a client



Review (cont'd)

The source is the packet sender

The destination is the packet receiver

In a network conversation, nodes take turns being source and destination

The OSI model describes the 7 levels of networking, and the protocols at each level (layer)

The TCP/IP model has 4 layers; the layers map to corresponding OSI layers

The TCP/IP model has six core protocols: TCP, UDP, IP, ICMP, IGMP, ARP

TCP is connection-oriented and “reliable”, and is used when completeness of data transfer is more important than the time it takes

UDP is connectionless, offers “best effort” service, and is used when time is more important than loss



Review (cont'd)

Both TCP and UDP embed port numbers in their headers

IP is used to carry a packet to its final destination

ICMP is used by IP-enabled nodes to send error and status messages to each other

IGMP is used by end devices to tell the router they still wish to receive a particular multicast

ARP is used to discover and map MAC addresses to IP addresses



Review (cont'd)

A router makes forwarding decisions based on Layer 3 (IP) addresses

A switch makes forwarding decisions based on Layer 2 (MAC) addresses

Layer 3 switches also have routing capabilities

Next generation firewalls and Intrusion Prevention Systems offer advanced features

Access points connect wireless devices to the wired LAN

Wireless controllers centrally manage multiple access points



Review (cont'd)

An endpoint is any device that connects to a network to communicate with other endpoints use their services

Power over ethernet allows DC power to be carried along a copper cable to the end point.

Traditional Cisco topology is a three-tier model consisting of a core layer, distribution layer, and access layer.

In smaller environments, the core and distribution layers can be combined (aka two-tier or collapsed core)

A Spine-Leaf topology uses a type of collapsed core to allow Leaf nodes (servers, etc.) to connect directly to the spine



Review (cont'd)

WAN topologies include point-to-point, Hub-and-Spoke (AKA star), partial mesh, full mesh, and dual-homed

6-to-4 tunneling allows an IPv6 logical topology to be overlaid on top of an IPv4 physical topology

A Small Office Home Office (SOHO) usually consists of multiple users sharing a single broadband connection through one router

A corporate Network can consist of any combination of on-premises and cloud topologies, including separate datacenters, branch offices, and remote workers



Review (cont'd)

Ethernet is the most common type of layer 2 protocol. It is transmitted over wired connections

Point to point connections involve two devices directly connected. They can include straight through connections between computer and switch, or crossover connections between two switches (or like devices).

A shared media connection is one in which multiple nodes are either physically or logically plugged into the same conductor

The key point of any shared media connection is that when one node transmits, all other nodes hear the transmission

There are a variety of Ethernet over copper standards, ranging in speed and distance

Twisted pair cable categories currently range from Cat5e (gigabit Ethernet) to Cat 8 (40 gigabit Ethernet)

Coaxial cable is still used today in radio, TV, and cable modems

Twin ax is a variation on coax that is a lower cost alternative to fiber optic cable



Review (cont'd)

Ethernet can also be carried over fiber optic cable.

Fiber optic cable transmits light on a glass strand as opposed to electrical signals on a copper wire.

Fiber optic cable is more expensive than twisted pair, but it can carry transmissions much farther and is immune to electromagnetic interference

Fiber optic requires either two strands (one to transmit and one to receive) or two different frequencies on the same strand

Fiber optic cable comes in multimode and single mode.

Multi-mode fiber is cheaper, Has a much larger core, travels a shorter distance, and is suitable for connecting switches within the same building

Single mode fiber is more expensive, has a much smaller core, travels a much longer distance, and is suitable for connecting switches between buildings, or across geographical areas

It's a bad idea to mix single mode and multimode fiber optic cable along the same path



Review (cont'd)

Virtual machines are entire operating systems that run as a guest application inside of a host operating system

Virtual machines appear on the network as if they are separate physical computers with their own IP address, MAC address and services

It is common practice to virtualize many servers onto a powerful host computer to reduce space and power consumption in a data center

A container is a sort of virtual machine for a single application

Each container contains just the app and enough files (libraries) to run the app

Containers share the host operating system and Hardware

VRF is a technology that allows one router to have multiple routing tables

It can be thought of as the layer 3 equivalent of a VLAN

It is used by service providers to isolate different customers on the same hardware and network link

